Dependable and Secure Privacy Trading in Federated Learning via MFG-RegretNet

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***Abstract*—Federated learning (FL) keeps raw data local, but repeated model updates still accumulate privacy loss, which FL markets rarely price. Buying each client's differential-privacy (DP) budget compensates this loss, but such a market must elicit valuations and privacy budgets truthfully under a fixed payment budget, stay tractable for large populations, and choose which privacy to buy, since more privacy alone does not make the global model more accurate. We derive an analytical Privacy-Aware Auction Mechanism (PAC) that is dominant-strategy incentive compatible (DSIC), individually rational (IR), and budget-feasible. PAC converges to its closed-form mean-field limit at rate $\mathcal{O}_P(N^{-1/2})$. Under budget-weighted aggregation, we bound the global model's error by a data-alignment term plus a DP noise term, so purchases misaligned with the clients' data shares bias the model. Our learned mechanism, MFG-RegretNet, therefore sets how much privacy to buy from a mean-field statistic of the bids and splits it in proportion to the selected clients' data. Its monotone networks and Myerson payments make it DSIC, IR, and budget-feasible for every parameter value with verified data sizes, and with numerical quadrature, no misreport gains more than $\overline{v}\overline{\epsilon}/n$. In a pre-specified confirmation (18 settings, three fresh seeds, official test sets) with budget-weighted aggregation, MFG-RegretNet exceeds the best external mechanism of each setting on CIFAR-10 in every seed, by 1.70 points on average. Its measured regret stays below 10^{-5}, and an online attacker or a coalition within its caps gains at most about 1%.**

***Index Terms*—Federated Learning, Differential Privacy, Privacy Auction, Collusion-Resistant Mechanism Design, Mean-Field Game, Dependable Distributed Computing**

I. Introduction

FEDERATED Learning (FL) [1], [2], [3] has emerged as the predominant paradigm for privacy-aware distributed machine learning, keeping raw data on local devices while sharing only model gradients with a central server. Yet a fundamental tension persists: *service providers require high-quality data to build accurate models, while users are increasingly reluctant to disclose sensitive information without adequate compensation or privacy guarantees.* Beyond accuracy, large-scale FL deployments also require *dependability*: the incentive layer must remain computationally scalable, robust to strategic manipulation, and consistent with long-horizon privacy accounting when the same clients participate over many rounds. They also require *security*: privacy valuations and budget reports are sensitive market signals whose leakage or manipulation can affect both payments and privacy exposure [4]. Accordingly, the privacy-incentive layer must be evaluated as part of a secure distributed system: it should specify what information is visible to the broker, server, clients, and external observers; how bid confidentiality and gradient confidentiality can be protected; how strategic and coordinated manipulation are handled; and how cumulative privacy loss is accounted for across repeated FL rounds. This tension is exacerbated by two structural deficiencies that undermine practical FL adoption at scale.

First, FL does not eliminate privacy risk. Model gradients themselves constitute a serious privacy vulnerability: gradient inversion attacks [5] can reconstruct original training samples from uploaded gradients with high fidelity. Differentially private FL (DP-FL) [6] mitigates this risk by injecting calibrated noise into gradients before upload, but this introduces a new dilemma: different clients have heterogeneous privacy *valuations*: some are willing to upload relatively clean gradients (ϵ large) for higher compensation, while others require strong protection (ϵ small) and therefore contribute noisier, less useful gradients. A one-size-fits-all privacy policy fails to respect this heterogeneity, resulting in either privacy harm or avoidable accuracy loss.

Second, FL lacks effective incentive mechanisms for privacy-sensitive participants. In the absence of monetary compensation, rational users have little motivation to expend local computation, communication bandwidth, and crucially privacy budget. Most existing Federated Learning Market (FLM) approaches trade raw datasets, trained model weights, or FL training capacity [7], [8], [9]. Within FL, only a few works price privacy itself, notably the privacy–price contracts of Lin et al. [10] and the grouping-competition privacy allocation of Wu et al. [11]. Following this line, we treat the differential privacy budget ϵ_i, rather than data, models, or computation, as the *primary object of trade*, with users compensated in proportion to the privacy they sacrifice, and ask how such a market can be made truthful and budget-feasible, and how its analysis scales to large client populations.

Three Key Technical Challenges. Realizing this vision raises three fundamental challenges that prior work has not jointly addressed.

Challenge 1: Privacy Market Design. How should the privacy trading market be formally structured? A rigorous market design must specify: (i) how users express their heterogeneous privacy valuations; (ii) how a data buyer (e.g., an FL server)

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specifies its accuracy requirements and budget; and (iii) how a semi-trusted broker/auctioneer allocates privacy budgets and determines payments. Critically, the mechanism must satisfy *Dominant-Strategy Incentive Compatibility* (DSIC), so that truthful reporting of privacy valuations is each user's optimal strategy, and *Individual Rationality* (IR), so that participation is always at least as good as abstaining. Without these guarantees, rational users will strategically misreport valuations, corrupting both market efficiency and FL model quality.

Challenge 2: Scalability of Auction Equilibrium. Even with a well-defined mechanism, computing the Nash Equilibrium (NE) of a privacy auction game among N participants is intrinsically difficult. Each user's optimal bid depends on the strategies of all other $N-1$ participants, so the NE problem has dimension $\mathcal{O}(N^2)$ and is PPAD-hard in general [12]. For the large-scale FL settings of interest (N in the hundreds to thousands), exact NE computation is computationally intractable. *How can we approximate the auction equilibrium efficiently without losing theoretical guarantees?*

Challenge 3: Accuracy-Oriented Mechanism Design under Unknown Valuation Distributions. A mean-field view summarizes the population, but does *not* specify which privacy the server should buy. The server pays for an accurate model, yet buying more privacy does not by itself reduce the FL error: when updates are aggregated with weights proportional to the purchased budgets, a purchase that is not proportional to the clients' data biases the global update. Classical approaches (e.g., Myerson's optimal auction theory [13]) yield closed-form solutions only under restrictive distributional assumptions. In FL, privacy valuations are heterogeneous, correlated across rounds, and drawn from an *unknown* distribution. *How can a mechanism be learned that buys the privacy that most reduces the FL error while remaining exactly truthful, without assuming any closed-form prior over valuations?*

These challenges are addressed by a unified framework that combines a *privacy trading market*, a *mean-field game (MFG) approximation*, and a *deep learning-based auction solver*, termed ***MFG-RegretNet***. Building on privacy auctions [14] and differentiable mechanism design [15], it makes the following contributions to dependable and secure FL.

- ***Privacy Trading Market and Auction Game.*** A formal privacy trading market for FL is established, in which each client treats its personalized differential privacy budget ϵ_i as a tradable commodity and bids for compensation commensurate with its privacy cost $c(v_i, \epsilon_i) = v_i \cdot \epsilon_i$, and the server buys privacy under a total budget B. Unlike the privacy–price contracts of Lin et al. [10], designed for a known distribution of privacy-preference types without a hard procurement budget, the market is a budget-constrained reverse auction in which clients report both their privacy valuation and their privacy budget, paired with a threat model that separates incentive compatibility from bid confidentiality, broker trust, client collusion, and gradient leakage. The interaction is modeled as a *Privacy Auction Game* (PAG). Its analytical PAC procurement mechanism is DSIC, IR, and Budget Feasible (BF), its release does not depend on the declared privacy budget, and truthful reporting is a Bayesian-Nash Equilibrium (Theorem 1); Proposition 1 characterizes, for any mechanism, when truthful reporting of both the valuation and the privacy budget is optimal, and MFG-RegretNet is built to satisfy it.
- ***Mean-Field Game Approximation for Dependability.*** We model the population of FL clients as a *mean-field flow* $\mu_t \in \mathcal{P}(\mathcal{V} \times \mathcal{E})$ and reduce the multi-agent Nash problem to a coupled Hamilton-Jacobi-Bellman / Fokker-Planck system, in which each client responds to the population law instead of to the individual bids of all other clients (Theorems 2 and 3). For the analytical mechanism, the mean-field limit is computed in closed form, and the finite-N rule provably converges to it at rate $\mathcal{O}_P(N^{-1/2})$ (Theorem 4), as confirmed by simulations up to $N{=}10^4$; the incentive and accuracy results of the learned mechanism below use $N{=}10$ clients.
- ***MFG-RegretNet: Accuracy-Oriented Learned Auction.*** We show that, under budget-proportional aggregation, the FL objective $\mathcal{P}_2$ is bounded by a data-alignment term plus a DP noise term, and that for a fixed purchase the aligned release is the projection of the purchase onto the data shares (Lemma 1). MFG-RegretNet addresses Challenge 3 by learning, from the bids and the mean-field statistic b_{MFG}, how much privacy to buy in each round, and by splitting the purchase with this projection, without any closed-form prior over privacy valuations. Its networks are monotone, so the payment rule of Proposition 1 applies exactly: for every parameter value, MFG-RegretNet is DSIC over valuations and privacy budgets, IR, and budget-feasible, with verified data sizes (Proposition 2), and a data dividend returns part of the unspent budget to the clients. Training therefore needs no regret penalty, and with numerical quadrature the regret is bounded on every bid profile (Corollary 2).
- ***Security- and Dependability-Oriented Validation.*** A validation suite spans a pre-specified confirmation of downstream accuracy on fresh seeds and the official test sets, Gaussian RDP and Laplace cumulative accounting, coordinated collusion, and the time, memory, and secure-execution cost for up to 500 clients. The attacks include an online learning attacker and coalitions in six modes, among them cap overstatement outside the guarantee. In the confirmation, with budget-weighted aggregation, MFG-RegretNet exceeds the best external mechanism of each setting on CIFAR-10 in every seed (by 1.70 points on average) and is on par with it on MNIST (+0.04 points); a registered control attributes its gain over a uniform split of the same privacy to these weights.

The remainder of this paper is organized as follows. Section II and Section III present related work and background on differential privacy and federated learning, respectively. Section IV and Section V present the privacy trading market and auction game, and the PAG-FL framework, respectively. Section VI and Section VII present the MFG approximation and MFG-RegretNet, respectively. Section VIII presents experimental results, and Section IX concludes the paper.

TABLE I
COMPARISON OF REPRESENTATIVE RELATED WORKS ACROSS KEY DESIGN DIMENSIONS

Work	FL	Priv.	Auc.	MFG	DL	DSIC
Ghosh & Roth [14]	N/A	✓	✓	N/A	N/A	✓
Fleischer & Lyu [16]	N/A	✓	✓	N/A	N/A	*Approx.*
Yang et al. [17]	✓	N/A	✓	N/A	N/A	✓
Zheng et al. [8]	✓	N/A	✓	N/A	✓	*Approx.*
Sun et al. [18]	✓	N/A	N/A	✓	N/A	N/A
Hu et al. [19]	✓	N/A	N/A	✓	N/A	N/A
Dütting et al. [15]	N/A	N/A	✓	N/A	✓	*Approx.*
Lin et al. [10]	✓	✓	N/A	N/A	N/A	✓†
Wu et al. [11]	✓	✓	✓	N/A	N/A	✓
MFG-RegretNet (Ours)	✓	✓	✓	✓	✓	✓‡

✓ = supported, N/A = not supported, *Approx.* = approximate (augmented Lagrangian training). Priv.: privacy as commodity; Auc.: auction; MFG: mean-field scalability; DL: deep learning; † by contract self-selection; ‡ exact up to the payment quadrature (Corollary 2).

II. RELATED WORK

This study lies at the intersection of four research streams, summarized in Table I.

Privacy-preserving FL. Gradient inversion attacks [5], [20], [21] demonstrate that an honest-but-curious server can reconstruct raw samples from shared gradients. The dominant defense is Differential Privacy: DP-SGD and its federated variants [6], [22] inject calibrated Gaussian noise into gradients before upload, offering (ϵ, δ)-DP at the cost of model utility. Secure aggregation [23], [2] uses cryptographic protocols so that the server observes only the aggregate. These approaches treat ϵ as a fixed system parameter rather than as a user-controlled commodity tradable for compensation.

FL incentive mechanisms. Incentive design for FL has been studied via Stackelberg games [24], [25], MFG-based reputation auctions [18], matching games [26], and DRL-based pricing [27]. Auction-based FL (AFL) [28] has emerged as a prominent sub-field, with CSRA [17] providing robust reverse auctions for DP-FL with truthfulness and budget-feasibility guarantees. Most AFL works treat privacy as a cost constraint and trade data, models, or compute capacity rather than the privacy budget itself; the exceptions are discussed next.

Privacy auctions and data markets. Ghosh and Roth [14] laid the theoretical foundation for privacy-as-commodity by formulating a reverse auction that elicits individual privacy valuations. Fleischer and Lyu [16] extended this to correlated privacy costs; Ghosh et al. [29] addressed unverifiable valuations. More recently, Hu et al. [19] proposed MFG-based data pricing for federated markets. Two FL works are closest to ours. Lin et al. [10] bound the effect of heterogeneous DP noise on FL convergence and derive optimal privacy–price contracts for discrete and continuous privacy-preference types, so that clients truthfully reveal their privacy attitudes over multi-round training. Wu et al. [11] allocate privacy budgets in FL through a grouping competition and treat privacy as a truthfully reported resource. Our framework differs from both in three respects: (i) PAC is a budget-feasible reverse auction that is truthful in dominant strategies and whose release does not depend on the declared budget, whereas the contracts of [10] are optimized for a known type distribution without a hard procurement budget; (ii) MFG-RegretNet learns how much privacy to buy without a closed-form valuation prior, targets a bound on the FL error rather than revenue, and remains exactly truthful because its payments follow from a monotone allocation; and (iii) the learned mechanism reads the population through a mean-field statistic of fixed dimension. Most other work in this line studies static, single-round, centralized settings.

Differentiable mechanism design. Myerson [13] gives revenue-optimal auctions under restrictive distributional assumptions inapplicable to heterogeneous FL clients. RegretNet [15] parameterizes allocation and payment rules as neural networks trained to minimize ex-post regret, circumventing distributional priors; DM-RegretNet [8] applied this to FL model trading. MFG-RegretNet builds on RegretNet: it adds the mean-field statistic to the input and trades privacy budgets as the commodity, but instead of penalizing regret it restricts the network to monotone allocations and derives the payments from them [13], so DSIC holds for every parameter value; regret is then only measured, by a search over misreports of the valuation, the privacy budget, and the data size.

Analytical DP-FL selection and pricing. Recent analytical approaches address heterogeneous-ϵ selection and compensation directly. Privacy-aware client selection [30] and heterogeneous DP aggregation [31] optimize selection probabilities under non-uniform noise variance, but provide no economic mechanism or incentive-compatibility guarantees. Jointly-optimal selection-and-allocation mechanisms (JSAM-style, e.g., [32]) derive structural results (threshold selection, $\epsilon_k^* \propto p_k^{2/3} v_k^{-1/3}$) for a Bayesian-optimal design, which assumes that the distributions of the clients' types are known. Our key distinctions are: (i) a *distribution-free* learned purchase rule that is exactly truthful; (ii) an objective derived from the FL aggregation error, which aligns the purchase with the clients' data; and (iii) a bound on the regret of the deployed network that holds on every bid profile.

III. PRELIMINARIES

A. Differential Privacy

We adopt *Personalized Local Differential Privacy* (PLDP) [33], [34], which generalizes standard LDP [35] to allow each client to hold a distinct privacy budget.

Definition 1 (PLDP [33]) *For client $i \in \mathcal{N}$ with $\epsilon_i > 0$ and $\delta_i \geq 0$, a randomized mechanism $\mathcal{M}^i$ is (ϵ_i, δ_i)-PLDP if for all adjacent inputs $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}^i$ and all measurable sets $S \subseteq \mathcal{Y}^i$:*

$$\Pr[\mathcal{M}^i(\boldsymbol{x}) \in S] \leq e^{\epsilon_i} \Pr[\mathcal{M}^i(\boldsymbol{x}') \in S] + \delta_i. \tag{1}$$

Smaller ϵ_i implies stronger privacy at the cost of lower data utility.

In practice, private gradients are perturbed via the Gaussian Mechanism [35]: for $\epsilon_i < 1$, noise scale $\sigma_i \geq \Delta f \sqrt{2\ln(1.25/\delta_i)}/\epsilon_i$ makes the perturbed gradient (ϵ_i, δ_i)-DP, where Δf is the ℓ_2-sensitivity of the gradient function;

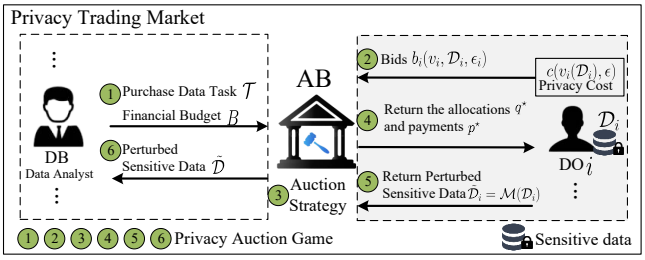


Fig. 1. Privacy Trading Market: the six-step auction process among DOs, the Auction Broker (AB), and DBs.

for larger budgets, σ_i is calibrated numerically from the Rényi-DP curve of the mechanism [36], as in our experiments. Either way, the allocated budget ϵ_i^{out} determines the noise injected into each client's update (see Algorithm 2).

B. Federated Learning

In standard FL, a server S coordinates N clients, each holding a private dataset $\mathcal{D}_i$. Client i minimizes its local loss $\mathcal{L}_i(\mathbf{w}, \mathcal{D}_i)$ to obtain local model $\mathbf{w}_i$; the server aggregates:

$$\mathbf{w}_G = \sum_{i=1}^{N} \alpha_i \mathbf{w}_i, \quad \sum_{i=1}^{N} \alpha_i = 1, \tag{2}$$

targeting $\mathbf{w}_G^* = \arg\min_{\mathbf{w}} \sum_i \alpha_i \mathcal{L}_i(\mathbf{w}_i)$. Under DP-FL, each client perturbs its gradient with Gaussian noise calibrated to ϵ_i^{out} before upload, so the server can only observe the noisy aggregate. The ℓ_2-sensitivity of client i's gradient is:

$$\Delta f_{\mathbf{w}_i} = \max_{\mathcal{D}_i, \mathcal{D}_i'} \|\nabla \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i) - \nabla \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i')\|_2 , \tag{3}$$

where $\mathcal{D}_i'$ differs from $\mathcal{D}_i$ in exactly one sample. The allocated budget ϵ_i^{out} thus determines the noise scale σ_i and directly affects gradient quality and downstream model accuracy, making it the natural economic object to trade in the privacy market.

IV. Privacy Trading Market and Privacy Auction Game

In this section, we introduce the *privacy trading market*, where privacy as a commodity trades among data owners (DOs) and data buyers (DBs). We then design a *privacy auction game* to incentivize participants to share data.

A. Privacy Trading Market

The privacy trading market comprises three roles: *Data Owners* (DOs), an *Auction Broker* (AB), and *Data Buyers* (DBs). As illustrated in Fig. 1, the auction proceeds in six steps. DOs $\mathcal{N} = \{1, \ldots, n\}$ hold local sensitive datasets $\mathcal{D} = \{\mathcal{D}_1, \ldots, \mathcal{D}_n\}$. We assume that DOs are *single-minded*: a DO will provide raw sensitive data only if the privacy preservation it requests is satisfied [37]. DO i protects her sensitive data through a mechanism $\mathcal{M}_{\epsilon_i}$ satisfying (ϵ_i, δ_i)-PLDP. DBs purchase sensitive data from DOs to complete analysis or model training tasks. The AB is a semi-trusted auctioneer that performs multiple rounds of auctions. Since a DO may misreport its privacy valuation $v(\mathcal{D})$, the mechanism must incentivize truthful reporting.

We model the AB as a semi-trusted auctioneer: it is expected to execute the published auction rule faithfully, but the bids it observes are privacy-sensitive and must be protected by protocol design in practical deployments.

Assumption 1 *All market participants agree to use privacy as a common evaluation metric, making the auction a fair game. Each DO knows the true value of its own data; v represents the DO's valuation of its data privacy during the auction. Each DO reports on its own behalf, so the formal guarantees concern unilateral deviations; coordinated misreports by coalitions are not excluded but evaluated in RQ1 (Sec. VIII-B1).*

Assumption 1 is the precondition for the formal incentive analysis of PAC and MFG-RegretNet. The threat model below characterizes a broader practical attack surface, including risks that are not covered by the exact DSIC/BF proof.

B. Security and Threat Model

Our mechanism operates at the auction layer and is orthogonal to the secure aggregation layer. The AB observes bid tuples such as (v_i, ϵ_i, d_i), the winner set, and payments. Data sizes enter the allocation and the dividend, so the AB verifies each client's data size when the client registers, for example through the data custodian or an attested count of local records; valuations and privacy budgets stay private and are reported strategically. The AB does not need raw data or individual plaintext gradients; these can remain protected by local DP [10] and secure aggregation protocols [23]. The FL server observes the aggregate model update and payment outcome, but not losing bids or individual valuations if the auction is executed inside a protected broker implementation; Supplementary Table S14[1] lists what each party sees.

We consider three security risks. First, an honest-but-curious AB may try to infer clients' privacy preferences from bids. DSIC addresses strategic misreporting, not bid confidentiality; preventing bid leakage requires a hardware-protected or cryptographic implementation, i.e., a trusted execution environment (TEE) or secure multi-party computation (SMPC). A practical TEE implementation proceeds as follows: the AB publishes an enclave measurement for the auction code; clients verify remote attestation and encrypt bids to the enclave; the enclave computes allocation and payments, then signs a commitment to the winner set and payment vector. An SMPC implementation can instead split bid processing across several non-colluding computation parties, using secure sorting and payment computation for PAC or an MPC-friendly neural forward pass for MFG-RegretNet, revealing only the final allocation and payment outputs; RQ2 measures a secure SMPC round and the channel cost of the TEE path. A malicious broker or FL server cannot weaken the DP guarantee, since each client adds its noise before upload, and in the TEE path the enclave attests the published network and signs the outcome, so each

[1]The supplementary material and the code are available at https://github.com/szpsunkk/MFG-RegretNet.

client can check that it is paid the signed amount and that $\sum_i \bar{p}_i \le B$. Second, clients may collude to manipulate bids or inflate payments. Our formal analysis assumes independent clients (Assumption 1); collusion-resistant auction design is outside the present theoretical guarantee. We therefore evaluate coordinated overbid, underbid, privacy-budget manipulation, and joint attacks empirically in RQ1. Third, gradient inversion is handled at the FL layer by DP noise and can be further mitigated by secure aggregation; it is not solved by incentive compatibility alone.

In deployment, the auction runs as a protected privacy-budget service: clients submit encrypted bids to a TEE- or SMPC-backed AB, which releases only the winners, payments, and budgets; the FL server runs DP-FL with secure aggregation; and an accountant records each client's cumulative exposure. PAC and MFG-RegretNet both give exact incentive guarantees; MFG-RegretNet adds a learned purchase rule that targets the FL error, and prices it by re-evaluating its allocation network on a valuation grid (Sec. VII). Under two-party SMPC, this re-evaluation dominates the cost of a round (RQ2).

C. Privacy Auction Game (PAG)

To incentivize privacy trading, we design a privacy auction game (Fig. 1). Each DO i has a valuation function $v_i(\mathcal{D}_i; b_i)$, where bidding profile b_i is given at privacy budget (ϵ_j, δ_j)-PLDP. DOs report their valuations (possibly untruthfully) and the AB decides an allocation and payment $p_1, \ldots, p_n$.

For DO i, the cost of having data used at privacy level ϵ is $c(v_i, \epsilon)$, linear in both v_i and ϵ: $c(v_i, \epsilon) \le c(v_j, \epsilon)$ if and only if $v_i \le v_j$. Specifically: $c_i(\epsilon) = \epsilon \cdot v_i$ [16].

The utility of DO i is:

$$u_i = p_i - c(v_i, \epsilon_i). \tag{4}$$

Summing client utilities defines *participant social welfare* (client-side economic surplus): monetary compensation net of privacy disutility. In one auction round,

$$\mathrm{SW} = \sum_{i=1}^{n} u_i = \sum_{i=1}^{n} \big(p_i - v_i \epsilon_i^{\mathrm{out}}\big), \tag{5}$$

where $\epsilon_i^{\mathrm{out}}$ is the privacy level actually applied to client i after allocation (in FL, the noise scale of the uploaded gradient). Under IR, $p_i \ge c(v_i, \epsilon_i^{\mathrm{out}})$ and each summand in Eq. (5) is non-negative. Payments are transfers from the server's budget B, so SW does not include the server's utility from model quality; the latter is evaluated via downstream accuracy (RQ4, Sec. VIII). In experiments we evaluate Eq. (5) using each client's *true* v_i when forming $c(\cdot)$, so that welfare aligns with realized privacy cost and is comparable across mechanisms. Across T_{FL} federated rounds we report the time average $\overline{\mathrm{SW}} = \frac{1}{T_{\mathrm{FL}}} \sum_{t=1}^{T_{\mathrm{FL}}} \mathrm{SW}^{(t)}$, where $(p_i^{(t)}, \epsilon_i^{(t),\mathrm{out}})$ denote round-t outcomes. We write SW to distinguish welfare from the type distribution denoted $\mathcal{W}$ in Theorem 1.

Define bidder i's type as $\theta_i = (v_i, \epsilon_i) \in \Theta_i$, where v_i is the privacy valuation and ϵ_i is the declared privacy-budget requirement. Let $\Theta = \Theta_1 \times \cdots \times \Theta_n$ and θ_{-i} denote the joint type profile excluding agent i.

D. Privacy Auction Equilibrium

All bidders aim to maximize $u_i(v_i, b)$. To obtain a *Nash Equilibrium*, we design a *privacy procurement mechanism*.

Definition 2 (Privacy Procurement Mechanism) *A privacy procurement mechanism $\mathcal{M} = (q, p)$ consists of:*

- *an allocation function $q : \Theta \to \{0,1\}^n$;*
- *a payment function $p : \Theta \to \mathbb{R}^n$.*

The mechanism $\mathcal{M}$ must satisfy the following properties:

- **Two-Dimensional Dominant-Strategy Incentive Compatibility (DSIC):** For each bidder $i \in \mathcal{N}$ and $\forall \theta_{-i} \in \Theta_{-i}$, truthful reporting of both valuation and privacy-budget requirement, $\theta_i = (v_i, \epsilon_i)$, is a dominant strategy regardless of others' reports:

$$u_i(\theta_i, \theta_{-i}) \ge u_i\Big(\hat{\theta}_i, \theta_{-i}\Big), \quad \forall \hat{\theta}_i \in \Theta_i. \tag{6}$$

- **Individual Rationality (IR):** For each bidder $i \in \mathcal{N}$, participation yields non-negative utility:

$$p_i(v) \ge c(v_i, \epsilon_i). \tag{7}$$

- **Budget Feasibility (BF):**

$$\sum_{i=1}^{n} p_i(v_i) \le B. \tag{8}$$

Under DSIC and IR, the expected revenue given joint distribution $\mathcal{W}$ of (v, ϵ, δ) is:

$$\mathcal{R} := \mathbb{E}_{(v,\epsilon)\sim\mathcal{W}} \left[\sum_{i=1}^{n} p_i(v, \epsilon) \right]. \tag{9}$$

Revenue-oriented privacy auctions maximize the expected payment to the clients:

$$\mathcal{P}_1 : \quad \max_{(q,p)} \ \mathcal{R} \quad \text{s.t.: DSIC, IR, BF.} \tag{10}$$

With unconditional transfers, $\mathcal{R}$ alone is not a procurement objective: paying out the budget maximizes it without buying useful privacy. This paper therefore optimizes the accuracy-oriented problem $\tilde{\mathcal{P}}_2$ of Sec. V and reports $\mathcal{R}$ only as an efficiency metric.

Definition 3 (Privacy Auction Equilibrium) *A strategy profile $s = (s_1, \ldots, s_n)$, where $s_i : \Theta_i \to \Theta_i$ maps a type to a report, is a Bayesian-Nash Equilibrium (BNE) if, for all $i \in \mathcal{N}$, $\theta_i \in \Theta_i$, and $\theta_i' \in \Theta_i$:*

$$\mathbb{E}_{\theta_{-i}}\big[u_i(s_i(\theta_i), s_{-i}(\theta_{-i}); \theta_i)\big] \ge \mathbb{E}_{\theta_{-i}}\big[u_i(\theta_i', s_{-i}(\theta_{-i}); \theta_i)\big]. \tag{11}$$

Theorem 1 (Privacy Auction Game Equilibrium) *In a PAG where privacy valuations v_i of DOs are drawn from joint distribution $\mathcal{W}$, the cost function is linear $c(v_i, \epsilon_i) = \epsilon_i \cdot v_i$, and the mechanism $\mathcal{M} = (q, p)$ satisfies two-dimensional DSIC, IR, and BF, there exists a Bayesian-Nash Equilibrium in which all rational bidders truthfully report both their privacy valuations v_i* and *their privacy budgets ϵ_i: for all i and $\theta_i, \theta_i' \in \Theta_i$,*

$$\mathbb{E}_{\theta_{-i}}\big[u_i(\theta_i, \theta_{-i}; \theta_i)\big] \ge \mathbb{E}_{\theta_{-i}}\big[u_i(\theta_i', \theta_{-i}; \theta_i)\big]. \tag{12}$$

Algorithm 1: PAC Mechanism

Input: Privacy valuations v, datasets $\mathcal{D}$, budget B
Output: Perturbed data $\mathcal{D}'$, payments p

All participants submit privacy valuation v to AB;
Sort valuations: $v_1 \le v_2 \le \cdots \le v_n$;
Let k be the largest integer such that $c\Big(v_k,\, \frac{1}{n-k}\Big) \le \frac{B}{k}$;
for $i = 1$ **to** n **do**
  Compute perturbed data: $\mathcal{D}_i' \leftarrow \mathcal{D}_i + \mathcal{N}(0,\, \Delta_f^2 \sigma^2)$;
  if $i \le k$ **then**
    $p_i \leftarrow \min\Big(\frac{B}{k},\, c\Big(v_{k+1},\, \frac{1}{n-k}\Big)\Big)$;
  else
    $p_i \leftarrow 0$;
  end
end
return $\mathcal{D}'$, p;

Proofs of all formal results are in Supplementary Sec. S4.

Remark 1 (Scope of Incentive Guarantees) *Theorem 1 holds for every mechanism with two-dimensional DSIC, IR, and BF. The* PAC mechanism *(Algorithm 1), which adopts the selection and payment rule of FairQuery [14] for this market, satisfies them, with DSIC over the two-dimensional type (v_i, ϵ_i). Exact DSIC for an arbitrary neural payment rule requires monotonicity conditions that cannot be verified for unconstrained deep networks [15]; Proposition 1 states them for the present setting.* MFG-RegretNet *meets them by construction: its allocation network is monotone and its payments take the form of Proposition 1, so the theorem applies to it as well (Proposition 2). Its only approximation is the numerical quadrature of the payment, whose effect is bounded on every profile (Corollary 2) and measured in RQ1 (Section VIII).*

PAC selects the k *lowest-valuation clients* as winners (monotone allocation: adding a client can only increase k, not decrease it), assigns each winner $\epsilon_i^{\text{out}} = 1/(n-k)$ (equal-share DP budget), and pays each winner $p_i = \min(B/k,\, c(v_{k+1}, 1/(n-k)))$, i.e., a threshold payment equal to the critical bid at which the marginal client would be selected. The three properties follow directly: *(i) DSIC*: reporting $v_i' > v_i$ cannot increase p_i (threshold payment is determined by v_{k+1}, not v_i); misreporting $v_i' < v_i$ either leaves p_i unchanged (if i remains a winner) or yields $p_i = 0$ (if i drops out), so $u_i(v_i) \ge u_i(v_i')$. *(ii) IR*: $p_i = \min(B/k, c(v_{k+1}, \epsilon)) \ge c(v_k, \epsilon) \ge c(v_i, \epsilon)$ for $i \le k$ (since $v_i \le v_k \le v_{k+1}$). *(iii) BF*: $\sum_i p_i \le k \cdot B/k = B$. The choice $\epsilon = 1/(n-k)$ is a conservative uniform allocation rule: all selected clients use the same DP budget, which simplifies the threshold payment and avoids heterogeneous noise scales in aggregation. Since this release does not depend on the declared budgets, PAC applies to winners whose budget is at least $1/(n-k)$. The winner index k is chosen only when $c(v_k, \epsilon) \le B/k$, so the rule is compatible with the per-client affordability condition. More general non-uniform or learned privacy-budget rules are possible, but would require a separate monotonicity and critical-payment analysis to retain exact DSIC. *Complexity*: $\mathcal{O}(N \log N)$ (sorting).

The two-dimensional statement holds for any mechanism under a simple condition. Fix the other reports and the reported data size, and let $e(\hat{v}, \hat{\epsilon})$ and $p(\hat{v}, \hat{\epsilon})$ be client i's release and payment. A declared cap commits the client to accept any release up to it, so a single-minded client never declares more than its cap ϵ, but it may declare less; likewise, a client cannot report more data than it holds.

Proposition 1 (Truthfulness with One-Sided Caps) *Let $U(v, \hat{\epsilon}) = \int_v^{\bar{v}} e(s, \hat{\epsilon})\, ds + h(\hat{\epsilon})$. Truthful reporting is optimal for every type $(v, \epsilon) \in [0, \bar{v}] \times [0, \bar{\epsilon}]$ over all reports with $\hat{\epsilon} \le \epsilon$ if and only if, for some function h, (i) $e(\cdot, \hat{\epsilon})$ is non-increasing and $p(\hat{v}, \hat{\epsilon}) = \hat{v}\, e(\hat{v}, \hat{\epsilon}) + U(\hat{v}, \hat{\epsilon})$ for every $\hat{\epsilon}$, and (ii) $U(v, \cdot)$ is non-decreasing for every v.*

Condition (i) is Myerson's characterization [13] at each cap, and (ii) rules out profitable cap understatement (proof in Supplementary Sec. S4.B). PAC satisfies (ii) trivially, with equality, because its release $1/(n-k)$ ignores the declared cap. A rule whose payment ignores the cap while releasing $e = z\hat{\epsilon}$ violates (ii): the client gains $vz(\epsilon - \epsilon_{\text{floor}})$ by declaring the smallest admissible cap ϵ_{floor}.

V. Privacy-Aware Auction Federated Learning Mechanism

We combine the privacy auction game with FL to incentivize users to share data for three reasons. First, users are concerned about privacy risks from local model leakage. Second, without revenue, users have limited incentive to participate in FL training. Third, a unified privacy policy ignores user heterogeneity, reducing model accuracy. We propose the *Privacy Auction Game-based FL* (PAG-FL) framework (Fig. 2).

A. PAG-FL Framework

PAG-FL involves three entities:

1) *FL Server (FLS).* FLS publishes model training tasks (e.g., image classification, text translation) and purchases client gradients via the FL architecture.
2) *FL Broker (FLB).* FLB is the auction institution for user data privacy. It recruits FLCs eligible for a given task and runs the privacy auction to determine allocation and payment.
3) *FL Clients (FLC).* FLCs are users or organizations that provide raw training data. Each FLC trains locally, applies a DP mechanism, and uploads a private gradient to FLB for auction.

The operational steps are:

1) *Publish Task.* FLS publishes a learning task to FLB, which recruits eligible FLCs.
2) *Registration.* FLCs register with FLB and declare initial privacy requirements (ϵ, δ).
3) *Submit Bid.* Each FLC i submits a bid $b_i = (v_i, \epsilon_i, d_i)$, where v_i is a valuation, ϵ_i is the privacy budget, and $d_i = |\mathcal{D}_i|$ is the size of its local dataset.
4) *Privacy Payment.* FLB runs the auction (PAC mechanism, Algorithm 1) and FLS pays selected FLCs.
5) *Private Model.* Each selected FLC applies its personalized DP mechanism to the local gradient.

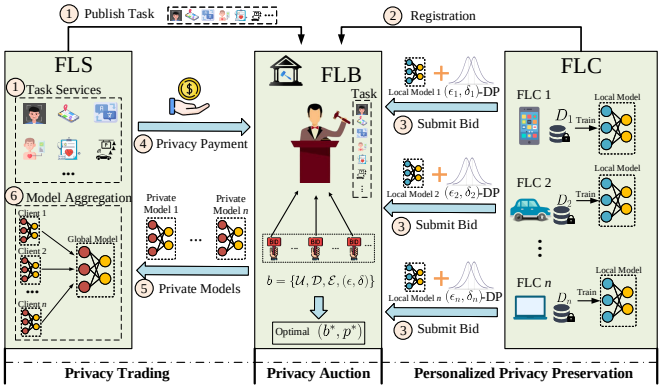


Fig. 2. The PAG-FL framework showing six operational steps between the FL Server (FLS), FL Broker (FLB), and FL Clients (FLC).

6) *Model Aggregation.* FLS aggregates received private gradients and broadcasts the updated global model.

B. FL Privacy Auction Model

1) Valuation Function of FLC: The valuation v_i of FLC i reflects the economic value that client i places on its privacy. It is determined by three factors: (i) the *sensitivity* of the local dataset $\mathcal{D}_i$, which is higher for medical or financial records than for generic browsing data; (ii) the *data quality*, quantified by the local model accuracy acc_i and dataset size $|\mathcal{D}_i|$; and (iii) the *non-IID degree* of the data distribution, measured by the divergence between the local and global data distribution. A concrete parameterization is:

$$v_i = \alpha_v \cdot \frac{\mathrm{acc}_i}{\epsilon_i} + \beta_v \cdot |\mathcal{D}_i| + \gamma_v \cdot \Delta_{\mathrm{noniid},i}, \tag{13}$$

where $\alpha_v, \beta_v, \gamma_v > 0$ are weighting coefficients and $\Delta_{\mathrm{noniid},i}$ quantifies the non-IID divergence (e.g. total variation between local and global class frequencies). This formulation captures the key observation that higher-quality and more privacy-sensitive data should command higher valuations. In the theoretical analysis (Sections VI through VII), v_i is treated as drawn from a generic distribution $\mathcal{W}$, so the results hold independently of the specific form of Eq. (13). In experiments (Section VIII), valuations are drawn directly from the scenarios of Supplementary Table S1: $v_i \sim \mathrm{Uniform}[0, 1]$, independent of ϵ_i, in the uniform and bimodal scenarios, and $v_i \sim \mathrm{LogNormal}(\mu{=}0, \sigma{=}0.5)$, divided by its 99th percentile and capped at $\bar{v}{=}1$, in the realistic scenario, reflecting heterogeneous real-world privacy sensitivities.

The cost $c_i(\epsilon_i^{\mathrm{out}}) = \epsilon_i^{\mathrm{out}} \cdot v_i$ depends on the *allocated* $\epsilon_i^{\mathrm{out}}$ (mechanism output), not the declared type ϵ_i; the privacy and cost trade-off is genuine.

Obtaining valuations in practice. The mechanism asks client i only for its own per-unit privacy price v_i; it needs neither the valuations of other clients nor the distribution $\mathcal{W}$. Because truthful reporting is a dominant strategy under PAC (Sec. IV-D) and under MFG-RegretNet (Proposition 2), a client that does not know how others bid gains nothing by shading its report, so its best response is its own subjective estimate. A deployment can set v_i in three ways. (i) Organizational clients such as hospitals or banks derive v_i from their data-governance policy, mapping the sensitivity class of $\mathcal{D}_i$ and the associated contractual or regulatory liability to a price per unit of ϵ. (ii) Individual users select one entry from a short menu of privacy levels with posted compensation in the client application, a stated-preference format used in empirical studies of privacy valuation [38], and the selected entry is converted into v_i. (iii) Otherwise, the client software computes a default from Eq. (13) using its own $|\mathcal{D}_i|$, acc_i, and $\Delta_{\mathrm{noniid},i}$. Since bids are re-submitted in every round (Algorithm 2), a client can revise v_i when its context changes, for example by raising it as its remaining privacy allowance (Sec. VIII-A1) shrinks. The surrounding protocol, i.e., encrypted bid submission to a protected broker, release of only the winners, payments, and budgets, DP-FL with secure aggregation, and per-client privacy accounting, is described in Sec. IV-B.

2) Bidding Information: In FL, global model accuracy depends on training dataset size, non-IID data distribution, and model privacy level [39], [40]. FLC i's bid is:

$$b_i = (v_i,\ \epsilon_i,\ d_i), \tag{14}$$

where ϵ_i and $d_i = |\mathcal{D}_i|$ indicate the privacy budget and the data size; the local accuracy acc_i enters only through the valuation, Eq. (13).

3) FL Aggregation Mechanism: Under privacy preservation, the aggregated global gradient error is:

$$\mathcal{L}(\mathbf{w}; \alpha, \epsilon) = \|\tilde{\mathbf{w}}_G - \mathbf{w}_G^*\|_\ell = \left\| \sum_{i=1}^{N} \alpha_i \mathcal{M}_{\epsilon_i}(\mathbf{w}_i) - \mathbf{w}_G^* \right\|_\ell . \tag{15}$$

The FL privacy auction objective is:

$$\mathcal{P}_2: \quad \min_{\mathbf{w}, \alpha, \epsilon} \mathbb{E}_{(\boldsymbol{b}, B)}[\mathcal{L}(\mathbf{w}; \alpha, \epsilon)] \quad \text{s.t.: DSIC, IR, BF.} \tag{16}$$

4) Economic Meaning of the Traded Budget: Following the privacy-auction literature [14], [16], the unit of trade is the DP parameter: in each round, client i sells the right to release its local update at privacy level $\epsilon_i^{\mathrm{out}}$ and is compensated for the resulting privacy cost $v_i \epsilon_i^{\mathrm{out}}$. For the server, the same number measures the quality of what it buys. If the local update is released with the classical Gaussian calibration of Sec. III, the per-coordinate noise variance is $\sigma_i^2 = 2\ln(1.25/\delta)\,\Delta f^2/(\epsilon_i^{\mathrm{out}})^2$, so a larger budget yields a less perturbed and thus more useful update. Algorithm 2 therefore weights the updates by their budgets, $\alpha_i = \epsilon_i^{\mathrm{out}} / \sum_{i' \in \mathcal{I}} \epsilon_{i'}^{\mathrm{out}}$: less perturbed updates count more, and a release in proportion to the data gives the winners their FedAvg weights. With independent zero-mean noise, the squared error of Eq. (15) with $\ell = 2$ decomposes as

$$\mathbb{E}\left\|\tilde{\mathbf{w}}_G - \mathbf{w}_G^*\right\|_2^2 = \left\|\mathbf{w}_G - \mathbf{w}_G^*\right\|_2^2 + \frac{C_{\mathrm{DP}}\,|\mathcal{I}|}{\left(\sum_{i \in \mathcal{I}} \epsilon_i^{\mathrm{out}}\right)^2}, \tag{17}$$

where $\mathcal{I}$ is the winner set, $\mathbf{w}_G = \sum_{i \in \mathcal{I}} \alpha_i \mathbf{w}_i$ is the noise-free aggregate, $C_{\mathrm{DP}} = 2 d_{\mathbf{w}} \Delta f^2 \ln(1.25/\delta)$ with $d_{\mathbf{w}}$ the number of model parameters, and $\epsilon_i^{\mathrm{out}} \geq \epsilon_{\min}$. The first term depends on data quality and heterogeneity, which d_i and acc_i describe; the second term is the DP noise that the server pays to reduce, and under these weights it depends only on the total purchase and the number of winners. A smaller budget thus gives a

client stronger privacy at the price of a noisier contribution, and the payment rule sets the exchange rate between the two: the noise term decreases with the square of the total purchased budget, while each client's privacy cost grows linearly in the budget it sells. With DP-SGD (RQ4), the noise enters during local training, and Eq. (17) describes the same trade-off qualitatively. Across rounds, the purchased budgets compose into each client's end-to-end guarantee as given in Eq. (41) of Sec. VIII-A1.

5) An Accuracy Bound for $\mathcal{P}_2$: The first term of Eq. (17) depends on which clients the weights favor. Let $\pi_i = d_i / \sum_{j\in\mathcal{N}} d_j$ be client i's share of the data and $\mathbf{w}_G^* = \sum_{i\in\mathcal{N}} \pi_i \mathbf{w}_i$ the full-data aggregate that FedAvg targets.

Assumption 2 (Bounded Heterogeneity) *The local updates satisfy $\sum_{i\in\mathcal{N}} \|\mathbf{w}_i - \mathbf{w}_G^*\|_2^2 \le \zeta^2$.*

Lemma 1 (Alignment Bound) *Under Assumption 2 and the weights $\alpha_i = \epsilon_i^{\text{out}}/S$ of Eq. (17), with $S = \sum_{i\in\mathcal{I}} \epsilon_i^{\text{out}}$ and $\alpha_i = 0$ for $i \notin \mathcal{I}$,*

$$\mathbb{E}\big\|\tilde{\mathbf{w}}_G - \mathbf{w}_G^*\big\|_2^2 \le C_{\text{DP}}\, J_\kappa(\boldsymbol{\epsilon}^{\text{out}}),$$
$$J_\kappa(\boldsymbol{\epsilon}^{\text{out}}) = \kappa\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2^2 + \frac{|\mathcal{I}|}{S^2}, \tag{18}$$

where $\kappa = \zeta^2/C_{\text{DP}}$. Moreover, among all releases with total S that respect the declared budgets, the Euclidean projection of $S\boldsymbol{\pi}$ onto $\{\mathbf{e} : 0 \le e_i \le \hat{\epsilon}_i,\ \sum_i e_i = S\}$ minimizes $\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2$.

Since both weight vectors sum to one, $\mathbf{w}_G - \mathbf{w}_G^* = \sum_i (\alpha_i - \pi_i)(\mathbf{w}_i - \mathbf{w}_G^*)$, and the Cauchy–Schwarz inequality bounds the first term of Eq. (17) by $\zeta^2\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2^2$; for a fixed total, $\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2 = \|\boldsymbol{\epsilon}^{\text{out}} - S\boldsymbol{\pi}\|_2/S$ (Supplementary Sec. S4.C). The weight κ sets the exchange rate between the two sources of error: a large κ (heterogeneous clients) favors a purchase in proportion to the data, and a small κ favors a larger purchase from fewer clients. Because ζ^2 is not observed in practice, the buyer publishes κ as a task parameter (Sec. VIII states how it is set). The bound turns $\mathcal{P}_2$ into a procurement objective that depends only on the reports and the release,

$$\tilde{\mathcal{P}}_2 : \quad \min_{(q,p)} \mathbb{E}_{(\boldsymbol{b},B)}\big[J_\kappa\big(\boldsymbol{\epsilon}^{\text{out}}\big)\big] \quad \text{s.t.: DSIC, IR, BF.} \tag{19}$$

With DP-SGD, the noise enters during local training, and Eq. (18) describes the trade-off qualitatively, as Eq. (17) does: the bound concerns one round of output perturbation relative to the data-weighted reference model, and κ is a hyperparameter set on development runs (Sec. VIII).

VI. Mean-Field Game Approximation for the Privacy Auction

In Sections IV and V, we formulated the PAG and the accuracy-oriented procurement problem $\tilde{\mathcal{P}}_2$ (Eq. (19)) over N players. Although Theorem 1 guarantees the existence of a Bayesian-Nash Equilibrium under the PAC mechanism (Algorithm 1), *computing* that equilibrium becomes computationally intractable as the number of FL clients N grows. This section introduces a Mean-Field Game (MFG) approximation in which each client responds to the population law rather than to the individual bids of the other $N-1$ clients. At the exact Mean-Field Equilibrium (MFE), the induced strategy profile is an $\mathcal{O}(N^{-1/2}\log N)$-Nash equilibrium of the N-player PAG (Theorem 3); this bound concerns the MFE itself, and Remark 2 states how the deployed MFG-RegretNet relates to it. For the analytical rule of Algorithm 1, the mean-field solution is computed in closed form and the deployed finite-N mechanism converges to it at rate $\mathcal{O}_P(N^{-1/2})$ (Theorem 4). We model FL rounds by a continuous time index $t \in [0, T]$ so that the HJB-FP formalism describes the *law* of valuations and privacy types; the auction *within* each round remains the static PAG of Sec. IV, embedded in the representative-agent utility Eq. (21).

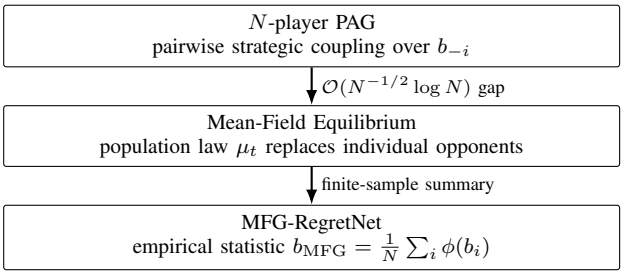


Fig. 3. Approximation hierarchy from the original N-player privacy auction to the deployable MFG-RegretNet statistic. The gap of Theorem 3 refers to the upper arrow. For the lower arrow, the deployed network is DSIC by construction (Proposition 2), so the statistic cannot be used for manipulation.

The intuition is simple. In the original N-player auction, each client's best response depends on the bids of all other $N-1$ clients. The MFG approximation replaces these pairwise dependencies with a population distribution μ_t. MFG-RegretNet then gives its networks the empirical mean of the encoded bids, b_{MFG}, as a finite-population summary of that distribution. The MFE approximates the true N-player equilibrium; the network is not trained to solve the MFE, and its incentive properties do not rely on it, because they hold by construction (Sec. VII-C).

A. Scalability and MFG State Space

For a mechanism without dominant strategies, the NE of the N-player PAG requires solving N coupled optimizations, which is PPAD-hard in general [12]; best-response iteration repeats an $\mathcal{O}(N\log N)$ evaluation of a sorting-based mechanism in each of $\mathcal{O}(N)$ iterations. (PAC needs none of this, since truthful reporting is dominant.) The MFG framework [41], [42] replaces explicit agent-to-agent coupling with a representative agent interacting with the aggregate population law μ_t (Corollary 1).

Each DO's state at round t is $\mathbf{s}_i(t) = (v_i(t), \epsilon_i(t)) \in \mathcal{S} \triangleq \mathcal{V} \times \mathcal{E}$, evolving via the controlled SDE:

$$d\mathbf{s}_i(t) = f\big(\mathbf{s}_i(t), b_i(t), \mu_t\big)\, dt + \sigma\, dW_i(t), \tag{20}$$

where $b_i(t) \in \mathcal{B}$ is the bid, f is the drift, $\sigma > 0$ is a noise coefficient, $\{W_i(t)\}$ are independent Brownian motions, and the state is reflected at the boundary of $\mathcal{S}$ (R6). The empirical distribution $\mu_t^N = \frac{1}{N}\sum_i \delta_{\mathbf{s}_i(t)}$ converges weakly to the mean-field flow μ_t under standard Lipschitz regularity conditions on f (Assumption 3).

Assumption 3 (Regularity Conditions) *(R1) Lipschitz drift:* $\|f(s,b,\mu) - f(s',b,\nu)\| \le L_f(\|s-s'\| + W_1(\mu,\nu))$.
(R2) Compact controls: $\mathcal{B} \subset \mathbb{R}^m$ *compact and convex.*
(R3) Bounded costs: $c(v,\epsilon) = v \cdot \epsilon$ *bounded and continuous, and* $\tilde{p}(b;\mu)$ *continuous in* b.
(R4) Finite-moment initial law: $\mu_0 = \mathcal{W} \in \mathcal{P}_2(\mathcal{S})$.
(R5) Exogenous types: bids do not affect the type dynamics, $f(s,b,\mu) = f(s,\mu)$*, and in each round the clients' types are i.i.d. draws from* μ_t.
(R6) Compact state space: $\mathcal{S}$ *is compact and convex, and the state is reflected at its boundary.*

B. Hamilton-Jacobi-Bellman and Fokker-Planck Equations

The full derivation is in Supplementary Sec. S3. In the mean-field regime, each representative agent maximizes cumulative *mean-field utility*

$$u^{\mathrm{MFG}}(s,b;\mu_t) = \tilde{p}(b;\mu_t) - c(v,\epsilon(b)), \tag{21}$$

where $\tilde{p}(b;\mu_t)$ is the mean-field payment, the payment for bid b against opponents whose types have law μ_t (Assumption 4). The *mean-field bid* summarizes the competitive environment:

$$b_{\mathrm{MFG}}(t) = \mathbb{E}_{\mu_t}[b] = \int_{\mathcal{S}} b(s;\mu_t)\, d\mu_t(s). \tag{22}$$

The optimal bid induced by the HJB Hamiltonian is

$$b^*(s,t;\mu) = \arg\max_{b\in\mathcal{B}} \left[\tilde{p}(b;\mu_t) - c(v,\epsilon(b)) + (\nabla_s \Phi)^\top f(s,b,\mu_t)\right]. \tag{23}$$

Under (R5) the last term does not depend on b, so b^* maximizes the round's mean-field utility Eq. (21), consistent with the static auction within each round. The value function Φ and the population density μ_t are jointly characterized by the coupled HJB-Fokker-Planck system (Supplementary Eqs. (S2) and (S3)); their fixed point defines the Mean-Field Equilibrium below.

C. Mean-Field Equilibrium: Definition and Existence

Definition 4 (Mean-Field Equilibrium (MFE)) *A pair* (Φ^*, μ^*) *is a MFE if:*

(i) Optimality: Given μ^*, Φ^* *solves the HJB equation, Eq. (S2), and* b^* *is the maximizer of Eq. (23);*
(ii) Consistency: μ^* *solves the FP equation, Eq. (S3), under* $b^*(\cdot\,;\mu^*)$*, with* $\mu_0^* = \mathcal{W}$.

Theorem 2 (Existence of MFE) *Under Assumption 3, the privacy auction MFG admits at least one Mean-Field Equilibrium* (Φ^*, μ^*).

Proof sketch. By (R5) the bids do not enter the type dynamics, so the population law solves a McKean–Vlasov Fokker–Planck equation, which has a unique solution under (R1) and (R6). Given this law, a measurable maximizer of the Hamiltonian exists by (R2) and (R3), and the value function solves the then linear HJB equation. The full proof is in Supplementary Sec. S4.D.

D. MFG Approximation Guarantee

Assumption 4 (Anonymous Lipschitz Payment) *The mechanism is anonymous: client* i*'s* N*-player payment is* $\tilde{p}(b_i;\mu_{-i}^N)$*, where* μ_{-i}^N *is the empirical law of the other clients' types, and* $\tilde{p}(b;\mu)$ *is* L_u*-Lipschitz in* μ *w.r.t.* W_1*:*

$$|\tilde{p}(b;\mu) - \tilde{p}(b;\nu)| \le L_u W_1(\mu,\nu), \quad \forall b \in \mathcal{B},\ \mu,\nu \in \mathcal{P}(\mathcal{S}). \tag{24}$$

This is an assumption on a smooth payment rule. The threshold rule of Algorithm 1 decides through an order statistic and is analyzed in Theorem 4 with the Kolmogorov distance.

Definition 5 (ε-Nash Equilibrium) *A strategy profile* $\mathbf{b}^N = (b_1^N, \ldots, b_N^N)$ *is an* ε*-Nash Equilibrium if, for all* $i \in \mathcal{N}$ *and* $\hat{b}_i \in \mathcal{B}$*:*

$$\mathbb{E}\Big[u_i(\hat{b}_i, b_{-i}^N) - u_i(b_i^N, b_{-i}^N)\Big] \le \varepsilon. \tag{25}$$

Theorem 3 (MFG Approximation of Nash Equilibrium) *Let Assumptions 3 and 4 hold, and let* (Φ^*, μ^*) *be a MFE. Define the MFE-induced strategy profile:*

$$b_i^{N,*} = b^*(\mathbf{s}_i, t; \mu^*), \quad i = 1, \ldots, N. \tag{26}$$

Then $\mathbf{b}^{N,*} = (b_1^{N,*}, \ldots, b_N^{N,*})$ *is an* ε_N*-Nash Equilibrium (Definition 5) of the PAG, where:*

$$\varepsilon_N = \frac{2L_u C \log N}{\sqrt{N-1}} = \mathcal{O}\left(\frac{\log N}{\sqrt{N}}\right), \tag{27}$$

and $C > 0$ *depends only on the diameter of the compact state space* $\mathcal{S} \subset \mathbb{R}^2$.

Proof sketch. The gain of a deviation splits into its gain in the mean-field game, at most zero because b^* is a best response to μ^*, and two errors from replacing the empirical law of the opponents by μ^*, which Assumption 4 bounds by $2L_u W_1(\mu_{-i}^N, \mu^*)$; the rate of W_1 for $N-1$ i.i.d. points in $\mathbb{R}^2$ gives ε_N. The full proof is in Supplementary Sec. S4.E. In words, as the number of clients grows, replacing explicit client-to-client strategic coupling with a mean-field population summary becomes increasingly accurate at rate $\log N/\sqrt{N}$. The logarithm comes from the two-dimensional state (v,ϵ): the empirical measure of N i.i.d. points in $\mathbb{R}^2$ approaches its law in W_1 at rate $N^{-1/2}\log N$ [43], and in general this rate cannot be improved to $N^{-1/2}$ [44]. The theorem is stated for the exact MFE profile $\mathbf{b}^{N,*}$; the incentive properties of the learned mechanism of Sec. VII do not rely on it (Remark 2).

Corollary 1 (Computational Complexity Reduction) *Once* (Φ^*, μ^*) *is computed at a cost independent of* N *(via the coupled HJB-FP system, Eqs. (S2) and (S3)), evaluating* $b^*(\mathbf{s}_i;\mu^*)$ *for each agent costs* $\mathcal{O}(1)$*, so the MFE profile of* N *clients costs* $\mathcal{O}(N)$*, against* $\mathcal{O}(N^2 \log N)$ *for best-response iteration with a mechanism without dominant strategies.*

E. Mean-Field Limit of the Analytical Mechanism

For the analytical rule of Algorithm 1, the mean-field solution is available in closed form. Because the rule decides through an order statistic, we measure the population error with the Kolmogorov distance $\|F_N - F\|_\infty$ between the empirical and population CDFs of the reported valuations.

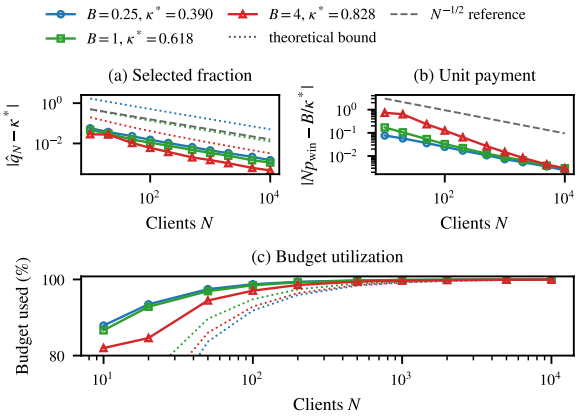


Fig. 4. Finite-N PAC against its mean-field limit (Theorem 4), $v \sim \mathrm{U}(0,1)$, 1024 markets per point, every simulated N plotted. (a) Selected fraction and (b) normalized per-winner payment; error bars are one standard error of the mean, and dotted lines in (a) are the bound $\sqrt{\pi/(2N)}/B + 1/N$ implied by Eq. (29). (c) Budget used; dotted lines are the deterministic lower bound of Eq. (29), averaged over markets. Analytical PAC mapping: the release floor $\epsilon_{\min}$ and the caps of the DP executor are not applied.

Theorem 4 (Mean-Field Limit of PAC) *Let $v_1, \dots, v_N$ be i.i.d. with CDF F on $[0, \bar{v}]$ and density $f \geq f_{\min} > 0$, let the budget $B > 0$ be fixed, and let $\kappa^* \in (0,1)$ be the unique root of*

$$\Psi(\kappa) = \kappa\big(F^{-1}(\kappa) + B\big) - B. \tag{28}$$

Let k_N be the number of winners of Algorithm 1 and $\hat{q}_N = k_N/N$. If $N \geq \max\{(1-\kappa^*)^{-1}, 1 + \bar{v}/B\}$, then

$$|\hat{q}_N - \kappa^*| \leq \frac{\|F_N - F\|_\infty}{B f_{\min}} + \frac{1}{N}, \quad B - \frac{B+\bar{v}}{N - k_N} \leq \sum_i p_i \leq B. \tag{29}$$

Since $\Pr(\|F_N - F\|_\infty > s) \leq 2e^{-2Ns^2}$ [45], $\hat{q}_N \to \kappa^*$ at rate $\mathcal{O}_P(N^{-1/2})$; the per-winner release and payment satisfy $N\epsilon_{\text{win}}^{\text{out}} \to (1-\kappa^*)^{-1}$ and $N p_{\text{win}} \to B/\kappa^*$ at the same rate; and the budget is exhausted at rate $\mathcal{O}(1/N)$.

Proof sketch. Algorithm 1 accepts k if and only if $\Psi_N(k/N) \leq 0$, where Ψ_N is Ψ with F replaced by F_N. A density bounded below turns $\|F_N - F\|_\infty$ into a uniform perturbation of Ψ, whose slope is at least B; at κ^* the two arguments of the PAC payment coincide. The full proof is in Supplementary Sec. S4.F.

Fig. 4 checks the theorem on the PAC implementation of Sec. VIII with $v \sim \mathrm{U}(0,1)$, three budgets, and 1024 markets for each $N \in [10, 10^4]$. Both inequalities of Eq. (29) hold in all 30,720 markets, and the fitted log–log slopes (-0.52 to -0.63 for the selected fraction, -0.51 to -0.85 for the unit payment, and -0.96 to -1.00 for the unused budget) match the predicted $N^{-1/2}$ and N^{-1} rates.

F. MFG-Reduced Privacy Auction Problem

Theorem 3 and Corollary 1 motivate summarizing the opponents by a population statistic, and Lemma 1 turns the FL objective $\mathcal{P}_2$ into the procurement objective J_κ of $\tilde{\mathcal{P}}_2$. Together they give the tractable MFG-reduced problem:

$$\mathcal{P}_3: \quad \min_{q,p,\mu^*} \mathbb{E}_{s\sim\mu_0^*}\big[J_\kappa\big(\boldsymbol{\epsilon}^{\text{out}}(\mathbf{b}^{N,*}; b_{\text{MFG}})\big)\big] \tag{30}$$

$$\text{subject to} \quad \text{FP, HJB, DSIC, IR, BF.}$$

$\mathcal{P}_3$ has three advantages over the N-player form of $\tilde{\mathcal{P}}_2$: (a) *reduced dimensionality*: opponent interaction is summarized by the low-dimensional statistic b_{MFG}; (b) *equilibrium foundation*: the exact MFE induces an $\mathcal{O}(N^{-1/2}\log N)$-Nash profile of the N-player PAG by Theorem 3; (c) *neural-network solvability*: gradient-based optimization is applicable, and Sec. VII enforces DSIC, IR, and BF through the architecture rather than through penalties.

Remark 2 (Approximation Hierarchy) *Theorem 3 guarantees that the MFE strategy $b^*(\cdot;\mu^*)$ is an ε_N-Nash Equilibrium* given the exact mean-field flow μ^* *(solved from the coupled HJB-FP system). MFG-RegretNet does not solve the HJB-FP system: its networks read the reports together with the empirical mean of the encoded bids, b_{MFG}, a finite-particle estimate of $\mathbb{E}_{\mu_t}[\phi(b)]$, and are trained to minimize the bound of Lemma 1 (Sec. VII-B). The theoretical $\mathcal{O}(N^{-1/2}\log N)$ guarantee therefore applies to the exact MFE solution of $\mathcal{P}_3$ and is not inherited by the trained network. The network does not need it for its incentive properties, which hold by construction (Sec. VII-C) and are measured in Sec. VIII-B1.*

VII. MFG-RegretNet: Deep Learning Auction Solver

Remark 2 identifies two layers: (i) the MFE ε_N-approximates the N-player Nash Equilibrium (Theorem 3), with $\varepsilon_N = \mathcal{O}(N^{-1/2}\log N)$; and (ii) computing the exact MFE requires solving the coupled HJB-FP system (Eqs. (S2) and (S3)), which is costly when $\mathcal{W}$ is unknown and must be run iteratively. MFG-RegretNet does not solve this system. It *parameterizes* the allocation of $\mathcal{P}_3$ as a neural network that reads the reports together with a mean-field statistic b_{MFG} and restricts this network to a monotone class. Proposition 1 then fixes the payments, so DSIC, IR, and BF hold for every parameter value instead of being penalized during training, and training only minimizes the bound of Lemma 1. It needs no closed-form distributional assumptions.

We propose **MFG-RegretNet**. Fig. 5 has two parts: **(A)** forward map (Sec. VII-A) with the mean-field statistic, the selection threshold, the purchase network, the aligned split, and the payment layer; **(B)** training objective (Sec. VII-B).

A. Architecture and Forward Mechanism

Given the reports $\hat{\mathbf{b}} = (\hat{b}_1, \dots, \hat{b}_N)$ with $\hat{b}_i = (\hat{v}_i, \hat{\epsilon}_i, \hat{d}_i)$, the budget B, and the public weight κ of Lemma 1, three modules apply in sequence. Unlike vanilla RegretNet [15], the population enters through b_{MFG}.

1) Module 1: Auction Header Network (AHN): Each report is rescaled to the features

$$\mathbf{f}_i = \Big(1 - \frac{\hat{v}_i}{\bar{v}},\ \frac{\hat{\epsilon}_i}{\bar{\epsilon}},\ \frac{\ln(\hat{d}_i/d_{\min})}{\ln(d_{\max}/d_{\min})}\Big), \tag{31}$$

each non-decreasing in what the client offers: a lower price, a larger budget, and more data. A per-client encoder ϕ_{θ_h}, a

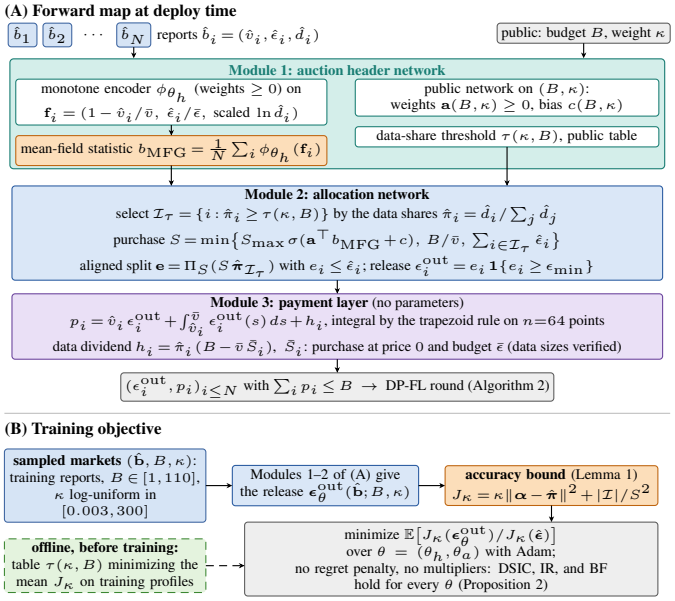


Fig. 5. MFG-RegretNet: (A) forward map at deploy time (Sec. VII-A); (B) training objective (Sec. VII-B).

tanh multilayer perceptron of width H whose weights are non-negative (softplus-parameterized), maps $\mathbf{f}_i$ to an embedding, and the AHN outputs the mean-field statistic

$$b_{\text{MFG}} = \frac{1}{N}\sum_{i=1}^{N} \phi_{\theta_h}(\mathbf{f}_i) \in \mathbb{R}^H, \tag{32}$$

the empirical mean of the encoded bids and a finite-particle estimate of $\mathbb{E}_{\mu_t}[\phi_{\theta_h}(b)]$; with the identity encoder it is the mean bid of Eq. (22). A public network reads only (B, κ) and outputs a weight vector $\mathbf{a}(B,\kappa) \in \mathbb{R}^H_{\geq 0}$ and a bias $c(B,\kappa)$, and a public table gives the selection threshold $\tau(\kappa, B)$ of Module 2. Every coordinate of b_{MFG} is non-decreasing in every feature of every client.

2) Module 2: Allocation Network: The allocation network first selects the clients whose reported data share $\hat{\pi}_i = \hat{d}_i / \sum_j \hat{d}_j$ reaches the public threshold, $\mathcal{I}_\tau = \{i : \hat{\pi}_i \geq \tau(\kappa, B)\}$ (the client with the largest share if no client does), then sets how much privacy to buy in the round,

$$S = \min\Big\{ S_{\max}\, \sigma\big(\mathbf{a}(B,\kappa)^\top b_{\text{MFG}} + c(B,\kappa)\big),\ \frac{B}{\bar{v}},\ \sum_{i\in\mathcal{I}_\tau} \hat{\epsilon}_i \Big\}, \tag{33}$$

and splits it in proportion to the data shares within $\mathcal{I}_\tau$:

$$\epsilon_i^{\text{out}} = e_i\, \mathbf{1}\{e_i \geq \epsilon_{\min}\}, \quad \mathbf{e} = \Pi_S\big(S\hat{\boldsymbol{\pi}}^\tau\big), \tag{34}$$

where $\hat{\pi}_i^\tau = \hat{\pi}_i / \sum_{j\in\mathcal{I}_\tau} \hat{\pi}_j$ for $i \in \mathcal{I}_\tau$ and 0 otherwise, and $\Pi_S(\mathbf{x})_i = \min\{\hat{\epsilon}_i^\tau, \max\{0, x_i + \mu\}\}$, with $\hat{\epsilon}_i^\tau = \hat{\epsilon}_i \mathbf{1}\{i \in \mathcal{I}_\tau\}$ and μ chosen so that the entries sum to S, is the Euclidean projection onto $\{\mathbf{e} : 0 \leq e_i \leq \hat{\epsilon}_i^\tau,\ \sum_i e_i = S\}$. Releases below the calibration floor $\epsilon_{\min}$ of Algorithm 2 are dropped, and no release exceeds a declared budget (allocation feasibility); dropping releases below $\epsilon_{\min}$ can lower the realized total below S. When every client is selected, the split is the minimizer of Lemma 1 for the total S; when some are excluded, that minimizer would project the full-population target $S\hat{\boldsymbol{\pi}}$, which shifts the selected entries by a constant instead of rescaling them, so the split is a proportional rule rather than the minimizer. The threshold trades the two terms of J_κ: a client with a very small share adds a release whose noise term $|\mathcal{I}|/S^2$ outweighs its alignment gain. The table $\tau(\kappa, B)$ is fixed before training (Sec. VII-B). When the buyer instead fixes the purchase T per round, as in the purchased-privacy settings of Sec. VIII, $S = \min\{T, \sum_{i\in\mathcal{I}_\tau} \hat{\epsilon}_i\}$ replaces Eq. (33), with a threshold table computed in the same way for fixed purchases; we require $T \leq B/\bar{v}$, which keeps (c) and (d) of Proposition 2.

3) Module 3: Payment Layer: Write $\epsilon_i^{\text{out}}(s)$ for client i's release when its valuation report is replaced by s and all other reports are fixed. The payment layer charges the payment of Proposition 1 with the term $h = h_i$:

$$p_i = \hat{v}_i\, \epsilon_i^{\text{out}}(\hat{v}_i) + \int_{\hat{v}_i}^{\bar{v}} \epsilon_i^{\text{out}}(s)\, ds + h_i, \quad h_i = \hat{\pi}_i\big(B - \bar{v}\,\bar{S}_i\big), \tag{35}$$

where $\bar{S}_i$ is the purchase of Eq. (33) when client i's price and budget are replaced by the most favorable ones, 0 and $\bar{\epsilon}$ (its data size is verified, Sec. IV-B). The first two terms pay for the release. The *data dividend* h_i returns part of the budget that the purchase leaves unspent, in proportion to the data shares, and is paid to every registered client, whether or not its release is executed: the purchase can only grow with a lower price or a larger budget (Proposition 2), so $S \leq \bar{S}_i \leq B/\bar{v}$ and $h_i \geq 0$, and h_i does not depend on client i's price or budget. The layer has no parameters of its own: it re-evaluates Module 2 on n equal intervals of $[\hat{v}_i, \bar{v}]$ and applies the trapezoid rule (n=64 in our experiments), and it evaluates $\bar{S}_i$ once. Only client i's embedding changes in these evaluations, so with the sum of Eq. (32) cached, a round costs $\mathcal{O}(nN(H^2 + N\log N))$, the projection included. No budget projection is needed: the first two terms are at most $\bar{v}\,\epsilon_i^{\text{out}}$, and $\bar{S}_i \geq S$ gives $\sum_i h_i \leq B - \bar{v}S$, so $\sum_i p_i \leq B$ for every report profile (Proposition 2).

Algorithm 2 summarizes one FL training round at deploy time: FLCs report bids (Sec. V), FLB forms the mean-field statistic, runs the MFG-RegretNet forward map (AHN, allocation, payment), and FLS aggregates DP private gradients weighted by the released privacy budgets.

B. Training Objective: the Accuracy Bound of $\mathcal{P}_2$

Because DSIC, IR, and BF hold for every parameter value (Sec. VII-C), training needs neither regret penalties nor Lagrange multipliers. It minimizes the bound of Lemma 1, the objective of $\mathcal{P}_3$, over sampled markets:

$$\mathcal{J}(\theta) = \mathbb{E}_{(\mathbf{b},B,\kappa)}\left[\frac{J_\kappa\big(\boldsymbol{\epsilon}_\theta^{\text{out}}(\mathbf{b}; B, \kappa)\big)}{J_\kappa\big(\hat{\boldsymbol{\epsilon}}\big)}\right], \tag{36}$$

where the denominator, the bound when every declared budget is bought, makes markets of different scales comparable, and the count $|\mathcal{I}|$ in J_κ is smoothed as $\sum_i e_i/(e_i + \epsilon_{\min})$. At a zero purchase this smoothed objective is finite, $\kappa\|\boldsymbol{\pi}\|^2$, whereas the exact bound is infinite, so for a very small κ it can favor buying nothing; RQ1 reports the share of markets without a purchase. Markets draw the reports from the training distribution, B over the budget range with half of the markets

Algorithm 2: MFG-Federated Privacy Auction: Online FL Round

Input: Trained $\theta^* = (\theta_h^*, \theta_a^*)$; global model $\mathbf{w}_G^{(t-1)}$; bids $\{b_i^{(t)}\}_{i=1}^N$ (Sec. V-B); budget B; weight κ; DP parameter δ; smallest budget $\epsilon_{\min} > 0$ that the local DP mechanism can calibrate
Output: Updated $\mathbf{w}_G^{(t)}$; payments $\{p_i^{(t)}\}$

$b_{\mathrm{MFG}}^{(t)} \leftarrow \frac{1}{N}\sum_{i=1}^N \phi_{\theta_h^*}(\mathbf{f}_i^{(t)})$;
$S^{(t)} \leftarrow$ Eq. (33); $\boldsymbol{\epsilon}^{\mathrm{out}} \leftarrow$ Eq. (34);
for $i = 1$ **to** N **do**
  $p_i^{(t)} \leftarrow$ Eq. (35) by the trapezoid rule on n intervals;
end
$\mathcal{I}^{(t)} \leftarrow \{i : \epsilon_i^{\mathrm{out}} \geq \epsilon_{\min}\}$ (the others do not upload);
for $i \in \mathcal{I}^{(t)}$ **do**
  From $\mathbf{w}_G^{(t-1)}$, train on $\mathcal{D}_i$ with a local DP mechanism calibrated to $(\epsilon_i^{\mathrm{out}}, \delta)$, e.g., DP-SGD, giving $\widetilde{\mathbf{w}}_i^{(t)}$;
end
$\alpha_i^{(t)} \leftarrow \epsilon_i^{\mathrm{out}} / \sum_{i' \in \mathcal{I}^{(t)}} \epsilon_{i'}^{\mathrm{out}}$ for $i \in \mathcal{I}^{(t)}$;
$\mathbf{w}_G^{(t)} \leftarrow \sum_{i \in \mathcal{I}^{(t)}} \alpha_i^{(t)} \widetilde{\mathbf{w}}_i^{(t)}$;
return $\mathbf{w}_G^{(t)}$, $\{p_i^{(t)}\}$;

at tight budgets, and κ log-uniformly over a range that covers the task weights, so one network serves every budget and task. The threshold table of Module 2 is fixed before training and is public: for each (κ, B) on a grid, $\tau(\kappa, B)$ is the threshold in $\{0, 0.005, \ldots, 0.25\}$ that minimizes the mean, over 4,096 training profiles, of the smallest J_κ that an aligned split among the selected clients attains, and it is interpolated in $(\log\kappa, \log B)$ between grid points. Supplementary Sec. S1.A gives the optimizer settings.

C. Formal Truthfulness Guarantees

Client i has type $t_i = (v_i, \epsilon_i, d_i)$ with $v_i \in [0, \bar{v}]$, $\epsilon_i \in [0, \bar{\epsilon}]$, $d_i \in [1, \bar{d}]$. Its data size is verified when it registers (Sec. IV-B), so it reports $\hat{b}_i = (\hat{v}_i, \hat{\epsilon}_i, d_i)$ with any $(\hat{v}_i, \hat{\epsilon}_i) \in \mathcal{B}_i(t_i) = [0, \bar{v}] \times [0, \epsilon_i]$, since caps cannot be overstated (Proposition 1): a release above a client's true cap is neither executed nor paid at settlement, and overstated caps lie outside the guarantee (RQ1 measures them). Its utility is $u_i = p_i - v_i\epsilon_i^{\mathrm{out}}$. DSIC is measured by *ex-post regret*:

$$\mathrm{rgt}_i(\theta; \mathbf{t}) = \max_{\hat{b}_i \in \mathcal{B}_i(t_i)} u_i(\hat{b}_i, \mathbf{t}_{-i}; \theta) - u_i(t_i, \mathbf{t}_{-i}; \theta), \tag{37}$$

where the network recomputes b_{MFG} from the deviating profile, and $r_i(\theta) = \mathbb{E}_{\mathbf{t}\sim\mathcal{W}}[\mathrm{rgt}_i]$.

Definition 6 (ε-Approximate DSIC) *The mechanism $\mathcal{M}_\theta$ is ε-approximately DSIC in expected ex-post regret if $r_i(\theta) \leq \varepsilon$ for all $i \in \mathcal{N}$.*

Proposition 2 (Exact Incentive Properties) *For every parameter value θ, budget B, and weight κ, with the integral of Eq. (35) evaluated exactly: (a) S and $\epsilon_i^{\mathrm{out}}$ are non-increasing in $\hat{v}_i$ and non-decreasing in $\hat{\epsilon}_i$; (b) truthful reporting of (v_i, ϵ_i) is a dominant strategy over $\mathcal{B}_i(t_i)$, i.e., $\mathrm{rgt}_i \equiv 0$; (c) $u_i \geq h_i \geq 0$ under truthful reporting (IR); and (d) $\sum_i p_i \leq B$ for every report profile (BF).*

Part (a) holds because the encoder, the mean, the sigmoid of a non-negative combination, and the minimum are monotone, the selection depends only on the verified data sizes, and the projection gives each selected client a release that is non-decreasing in its own budget and the total. Part (b) is the sufficiency direction of Proposition 1 with $h = h_i$, which does not depend on client i's report; (c) and (d) follow from $0 \leq \int_{\hat{v}_i}^{\bar{v}} \epsilon_i^{\mathrm{out}}(s)\, ds \leq (\bar{v} - \hat{v}_i)\, \epsilon_i^{\mathrm{out}}(\hat{v}_i)$ and $\bar{S}_i \geq S$. Proofs are in Supplementary Sec. S4.G.

Corollary 2 (Regret under Numerical Quadrature) *With the trapezoid rule on n equal intervals, (c) and (d) of Proposition 2 still hold exactly, and on every report profile $\mathrm{rgt}_i \leq \bar{v}\epsilon_i/n \leq \bar{v}\bar{\epsilon}/n$.*

The bound holds for every profile and every θ, without sampling; with $\bar{v}$=1, $\bar{\epsilon}$=5, and n=64 it is 0.078 per client and round, and RQ1 measures the regret actually attained.

Proposition 3 (From Regret to Truthfulness) *If $r_i(\theta) \leq \varepsilon$, then (a) for every $\tau > 0$, $\Pr[\mathrm{rgt}_i \geq \tau] \leq \varepsilon/\tau$; (b) the interim gain of any misreport against truthful opponents is at most $\mathbb{E}[\mathrm{rgt}_i \mid t_i]$, so if all clients have $r_i \leq \varepsilon$, truthful reporting is an ex-ante ε-Bayes–Nash equilibrium; (c) misreporting the valuation, the cap, or both gains at most rgt_i pointwise; and (d) if types are drawn afresh in each round, independently of past outcomes, then, since the network keeps no state across rounds, a strategy that adapts to all past outcomes gains at most $\sum_{\tau \leq T} \mathrm{rgt}_i(\theta; \mathbf{t}^{(\tau)})$ over T rounds against truthful opponents, i.e., $T\, r_i(\theta)$ in expectation.*

With Corollary 2, Proposition 3 holds with $\varepsilon = \bar{v}\bar{\epsilon}/n$ for every client and every θ.

MFG-Aware Regret Computation. To measure regret, the deviating report also moves the mean-field input:

$$b'_{\mathrm{MFG}} = b_{\mathrm{MFG}} + \frac{\phi_{\theta_h}(\mathbf{f}'_i) - \phi_{\theta_h}(\mathbf{f}_i)}{N}, \tag{38}$$

reflecting the agent's impact on the population-level statistic. We approximate the inner maximum by R steps of projected gradient ascent (PGA):

$$b_i'^{(r+1)} = \Pi_{\mathcal{B}_i}\left(b_i'^{(r)} + \eta_r \nabla_{b'_i} u_i(b'_i, b_{-i}; \theta)\right), \quad r = 0, \ldots, R-1, \tag{39}$$

initialized at the best point of a grid over $\mathcal{B}_i$ in $(v'_i, \epsilon'_i, d'_i)$ that includes ϵ'_i=0 (withholding). This yields the MFG-aware regret estimator:

$$\widehat{rgt}_i^{\mathsf{MFG}}(\theta) = \frac{1}{L}\sum_{l=1}^{L}\left[u_i(b_i'^{(R,l)}, b_{-i}^{(l)}; b_{\mathrm{MFG}}'^{(R,l)}, \theta) - u_i(b_i^{(l)}, b_{-i}^{(l)}; b_{\mathrm{MFG}}^{(l)}, \theta)\right]^+, \tag{40}$$

which a search can only underestimate; Corollary 2 gives the matching upper bound.

VIII. Performance Evaluation

We evaluate MFG-RegretNet through four research questions (RQs):

1) *RQ1 (Incentive Security):* Is MFG-RegretNet's measured regret within the bound of Corollary 2, with no IR

violations, and does its incentive stability persist under adaptive and coordinated bid manipulation?
2) *RQ2 (Dependable Scalability):* How do latency and memory, including the payment layer, scale with the client population size N, in plaintext and with the bids protected by secure computation?
3) *RQ3 (Privacy-Accounted Auction Efficiency):* How much privacy does each mechanism buy for its payment, at what participant welfare $\overline{\mathrm{SW}}$, and with what long-horizon privacy exposure?
4) *RQ4 (End-to-End FL Utility):* Under the same budget, or the same purchased privacy, is the model trained with the privacy that MFG-RegretNet buys more accurate than with that of the comparators, in a pre-specified confirmation on fresh seeds?

A. Experimental Setup

Data and models. On MNIST [46] we train a CNN of two convolutional and two fully connected layers, and on CIFAR-10 [47] a lightweight CNN of three convolutional blocks, a controlled FL benchmark rather than a state-of-the-art classifier; both datasets are partitioned among the clients by a Dirichlet distribution with $\alpha \in \{0.1, 0.5\}$.

Mechanisms. RQ1, RQ3, and RQ4 use $N{=}10$ clients and one privacy item, as in the RegretNet benchmarks [15], and RQ2 varies N up to 500. Auction profiles draw $v_i \sim \mathrm{Uniform}[0,1]$ ($\bar{v}{=}1$), $\epsilon_i \sim \mathrm{Uniform}[0.1,5]$ ($\bar{\epsilon}{=}5$), and data sizes from $\{10,\dots,500\}$; in RQ4 the sizes are the record counts of the partition, rescaled so that the largest is 500, with a floor of 10 that binds in no confirmation partition, so the shares equal the true data shares. MFG-RegretNet ($H{=}64$, $n{=}64$) is trained once, with seed 42, on Eq. (36) over $B \in [1,110]$ and $\kappa \in [0.003, 300]$, and one network serves every budget and task; its incentive properties hold for every parameter value (Proposition 2), so they do not depend on the training seed. The buyer sets $\kappa{=}3$ for MNIST and $\kappa{=}30$ for CIFAR-10, chosen among $\{0.01, \hat{\kappa}, 3, 30\}$, with $\hat{\kappa}$ a plug-in estimate of ζ^2/C_{DP}, by the development accuracy of the corresponding non-strategic aligned allocation on separate seeds, before any confirmation run. The code and the supplementary material, with the training and FL settings, the adaptation of every baseline, and the complete results of each RQ, are available online (footnote 1).

Baselines. PAC (Algorithm 1), VCG [48] in reverse-procurement form, RegretNet [15] without b_{MFG}, DM-RegretNet [8] (at epoch 150, before its training diverges), MFG-Pricing [19], CSRA [17] in one-shot adaptations without its reputation history (Sec. VIII-B1), a posted-price mechanism, and, in RQ4, Lin et al. [10] and PAC and VCG with each release lowered to the winner's declared budget.

1) Privacy Composition Across Rounds: The RDP costs of a client's releases add up over the set $\mathcal{H}_i$ of rounds in which it is selected [36], [49]. For Gaussian output perturbation calibrated to $\epsilon_i^{(t),\mathrm{out}}$ as in Sec. III, this gives

$$\epsilon_i^{\mathrm{tot}} \le \Gamma_i + 2\sqrt{\Gamma_i \log(1/\delta')}, \qquad \Gamma_i = \frac{\sum_{t\in\mathcal{H}_i} \left(\epsilon_i^{(t),\mathrm{out}}\right)^2}{4\ln(1.25/\delta)}, \tag{41}$$

with per-round parameter δ and end-to-end target δ'; Laplace noise composes by the sum of the budgets [50], and DP-SGD (RQ4) by the RDP curve of the subsampled Gaussian mechanism. Because noise is added before upload, the guarantee holds against the FL server and every party that observes the updates; it is record-level and does not hide whether a client participates. Before each release, the accountant, the same for every mechanism, excludes a client whose release would take it above its cap E_i^{tot}.

TABLE II
RQ1 AT THE BINDING BUDGET $B{=}25$ (10,000 HELD-OUT PROFILES): MEAN REGRET (GRID SEARCH PLUS PGA OVER $\mathcal{B}_i(t_i)$), SHARE OF OFFERED CONTRACTS WITH NEGATIVE UTILITY, AND BUDGET USE.

Mechanism	**Regret** (10^{-3})	**IR viol.**	$\mathcal{R}/B$
PAC / VCG	0	0	0.327
CSRA, release $1/(N{-}k)$	261	1.5%	0.734
CSRA, declared release	71.0	0	0.734
MFG-Pricing	22.0	6.7%	1.000
DM-RegretNet	62.8±16.4	7.6%	1.000
RegretNet	1.75±0.12	0	0.999
MFG-RegretNet, $\kappa{=}3$	$5.4{\times}10^{-6}$	0	0.956
MFG-RegretNet, $\kappa{=}30$	$1.9{\times}10^{-6}$	0	0.968

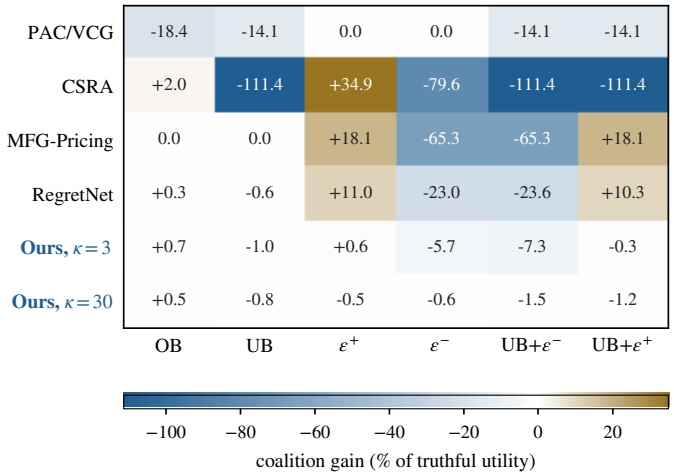


Fig. 6. RQ1: coalition gain under extreme coordinated attacks (% of the truthful utility; $B{=}50$, seeds 42–44, coalition ratios 0.1–0.5 pooled). OB/UB: over-/underbidding; ϵ^+/ϵ^-: cap inflation/deflation (ϵ^+ is outside the one-sided report set); CSRA: release $1/(N{-}k)$.

B. Experimental Results

1) RQ1: Incentive Compatibility: We measure the ex-post regret of Eq. (37) and the share of offered contracts with negative utility on 10,000 held-out profiles, with utilities under the true valuations (for MFG-RegretNet including the dividend). A client's best deviation comes from a $21{\times}21{\times}5$ grid over its valuation, its budget, and its data size, the last two up to the true ones, plus 25 PGA steps, and for MFG-RegretNet also from the sizes at every crossing of the selection threshold (analytical baselines: a $21{\times}21{\times}3$ grid). Sizes are verified, so size misreports test robustness beyond the guarantee. CSRA enters in two one-shot adaptations with the same winners and payments: in the first, used in RQ4, its winners release

TABLE III
RQ1: GAIN PER ROUND (% OF THE TRUTHFUL UTILITY) OF A CLIENT THAT LEARNS ITS MISREPORT ONLINE (B=25, 20,000 MARKETS, TEN RUNS).

Mechanism	After learning	Whole horizon	Best fixed policy	Best response
PAC	−0.24	−2.00	0.00	0.00
CSRA, release $1/(N-k)$	8.72	−12.10	11.60	13.40
MFG-Pricing	0.09	−12.70	0.31	1.80
RegretNet	−0.23	−6.33	0.10	0.13
DM-RegretNet	7.02	−3.74	7.75	10.28
MFG-RegretNet, κ=3	−1.13	−10.03	0.00	0.00
MFG-RegretNet, κ=30	−1.06	−8.36	0.00	0.00

TABLE IV
RQ2: ONLINE COST OF ONE SECURE ROUND (TWO-PARTY CRYPTEN, ONE SERVER; MEAN OF FIVE RUNS) AND THE TEE CHANNEL COST; TIME IN S.

	MFG-RegretNet, SMPC round			**Oblivious sort**			**TEE**
N	time	rounds	MB	time	rounds	MB	ms
10	3.28	606	106	0.50	90	0.04	0.027
50	7.96	804	1,730	1.03	189	0.31	0.062
100	37.58	930	7,602	1.44	252	0.83	0.105
200	165.04	1,074	35,249	1.86	324	2.14	0.193
500	1,096.13	1,236	215,286	2.37	405	5.35	0.456

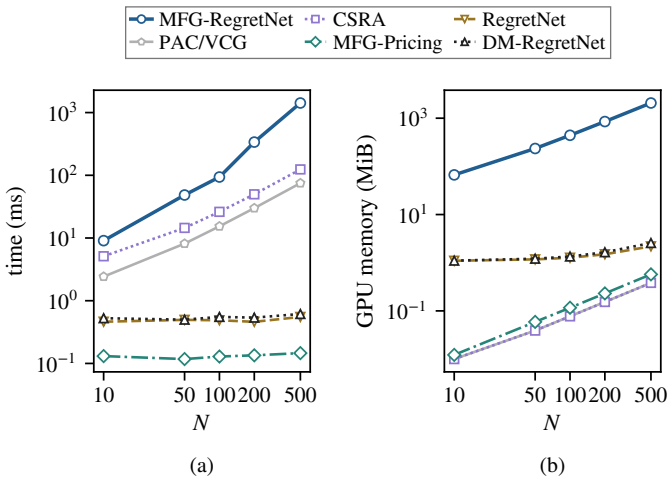


Fig. 7. RQ2: (a) time to clear a batch of 100 auction profiles; (b) incremental GPU memory of the auction step (one NVIDIA RTX A5000, median of seven interleaved calls; PAC, VCG, CSRA, and MFG-Pricing compute on the CPU).

$1/(N-k)$ as in PAC and VCG while their prices use the declared budgets, which causes its negative utilities; in the second, closer to the original rule, they release their declared budgets, which removes these and lowers the regret, but the larger releases lower its welfare (Tables II and V).

At the binding budget B=25 (Table II) and at the default B=50, the largest regret of MFG-RegretNet is 3.3×10^{-6}, far below the bound $\bar{v}\epsilon_i/n \geq 0.0016$ of Corollary 2; size misreports alone gain less than 10^{-12}, no offered contract has negative utility, and every market buys privacy. Its regret is far below that of the other learned and adapted mechanisms; only PAC and VCG, with closed-form threshold payments, have none. MFG-RegretNet pays 0.993–0.995 per unit of released privacy, close to the cap price $\bar{v}$.

Attacks. A client that learns its misreport online from its own outcomes, binning its type into eight contexts, trying each of 70 misreports of valuation, cap, and data size ten times per context, and then committing to the empirically best one (Table III), gains 8.7% of its truthful utility per round against CSRA and 7.0% against DM-RegretNet, but loses 1.1% against MFG-RegretNet, where no fixed misreport gains in hindsight; with Corollary 2, Proposition 3(d) bounds the gain of any adaptive strategy over T fresh markets by $T\bar{v}\bar{\epsilon}/n$. Proposition 2 covers unilateral deviations only, so we also measure coalitions of up to half of the clients (Fig. 6): within their caps they gain at most 1.0% under MFG-RegretNet, comparable to RegretNet (0.9%), against 7.6% under CSRA and 11.8% under PAC and VCG. Overstating the caps, outside the one-sided report set, gains at most 0.6% under MFG-RegretNet with κ=3 and nothing with κ=30, against 11.0–34.9% under RegretNet, MFG-Pricing, and CSRA; PAC and VCG ignore the caps.

2) RQ2: Scalability Analysis: Fig. 7 reports the time to clear a batch of 100 profiles and the incremental GPU memory, excluding local training. To price its release, MFG-RegretNet re-evaluates its allocation on n valuation values per client, at $\mathcal{O}(nN(H^2 + N\log N))$ per round (Sec. VII-A), and it is the slowest mechanism: it takes 9 ms at N=10 and 1.42 s at N=500, where it is 11–19 times slower than PAC, VCG, and CSRA; MFG-Pricing and the forward passes of RegretNet and DM-RegretNet, whose payments are network outputs, take below 1 ms. Its thresholds are set for N=10, so the figure reports cost only. With the bids secret-shared between two semi-honest CrypTen [51] servers (public weights and verified sizes; Table IV), a round takes 3.3 s and 106 MB at N=10 and 18.3 min and 215 GB at N=500: batching keeps the rounds within seven times those of one oblivious sort, the least that PAC, VCG, and CSRA need, but the traffic grows with the $(n+1)N$ projections of the payment layer, roughly as N^2. Under CrypTen's default fixed-point approximations, the secure releases and payments deviate from the plaintext ones by at most 0.03. A TEE broker adds 0.03–0.46 ms for bid decryption (AES-GCM) and outcome signing, plus about 14,000 cycles per enclave transition [52].

3) RQ3: Auction Efficiency: On the RQ1 profiles with truthful reports and $B \in \{25, 50, 75, 100\}$, we measure budget use $\mathcal{R}/B$, IR, and the participant welfare $\overline{\mathrm{SW}}$ of Eq. (5), the clients' surplus under their true valuations, which counts payments as income and so rises with every unit paid. MFG-RegretNet buys about 14.0 (κ=3) and 7.9 (κ=30) units of privacy per round at every budget, pays close to the cap price, and returns the rest of the budget through the dividend (Table V). Its $\overline{\mathrm{SW}}$ is the highest at every budget with κ=30 and from B=50 on with κ=3, but this margin is the dividend, a transfer that does not change the release. Over 200 rounds under Gaussian RDP (Fig. 8 shows the mean exposure), the first client of MFG-RegretNet reaches ϵ^{tot}=10 after 19–22 (κ=3) and 67–76 rounds (κ=30), against 4–10 under MFG-Pricing and RegretNet, which spend the whole budget, 75–77 under PAC and VCG, and 158–191 under CSRA; under Laplace composition, every mechanism reaches it within 19 rounds.

4) RQ4: Final Global Model Accuracy: RQ4 is a confirmation whose settings, seeds, comparators, endpoint, and decision rule were fixed before any of its runs. It runs the PAG-FL loop

TABLE V
RQ3: BUDGET USE, LARGEST SHARE OF CONTRACTS WITH NEGATIVE UTILITY, AND WELFARE $\overline{\mathrm{SW}}$ (10,000 ROUNDS PER BUDGET).

Method	$\mathcal{R}/B$	IR viol.	$\overline{\mathrm{SW}}$ at budget B 25	50	75	100
PAC / VCG	0.08–0.33	0	4.08	4.08	4.08	4.08
CSRA, release $1/(N-k)$	0.51–0.73	2.6%	17.14	29.56	39.56	47.50
CSRA, declared release	0.51–0.73	0	10.98	22.65	32.60	40.60
MFG-Pricing	1.00	6.7%	12.23	37.23	62.23	87.23
DM-RegretNet	1.00	7.6%	12.23	37.23	62.23	87.23
RegretNet	1.00	0	19.92	39.47	62.69	87.34
MFG-RegretNet, κ=3	0.96–0.99	0	16.93	41.89	66.92	91.99
MFG-RegretNet, κ=30	0.97–0.99	0	20.27	45.26	70.28	95.33

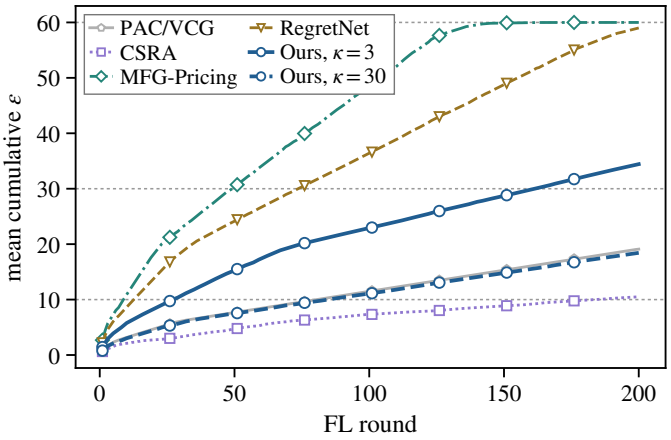


Fig. 8. RQ3: mean cumulative privacy ϵ^{tot} of the clients over 200 rounds under Gaussian RDP (B=50, three seeds), with the pre-release check at E_i^{tot}=60; dotted lines mark 10, 30, and 60.

TABLE VI
RQ4: FINAL ACCURACY $\mathcal{A}_{\mathrm{final}}$ (%) OF MFG-REGRETNET AND THE BEST EXTERNAL MECHANISM, AND MARGINS PER SEED (POINTS); THE DECISION USES ROWS "ALL".

Settings	$\mathcal{A}_{\mathrm{final}}$ (%) Ours	Best ext.	Margin (points) 231	232	233	Mean
MNIST						
budget B (6)	91.40	91.37	−0.67	+0.51	+0.27	+0.03
privacy $\bar{\epsilon}$ (4)	89.11	89.07	−1.42	+1.37	+0.17	+0.04
all (10)			−0.97	+0.85	+0.23	+0.04
Rule for MNIST: mean ≥ -0.3; met.						
CIFAR-10						
budget B (4)	31.84	30.52	+2.17	+0.34	+1.44	+1.32
privacy $\bar{\epsilon}$ (4)	32.02	29.94	+2.31	+2.36	+1.57	+2.08
all (8)			+2.24	+1.35	+1.51	+1.70
Rule for CIFAR-10: all seeds > 0; met.						

TABLE VII
RQ4 AGGREGATION CONTROL (CIFAR-10, B=30, DEVELOPMENT SEEDS): ACCURACY (%) WITH BUDGET (ϵ) OR DATA-SIZE (d) WEIGHTS. †ADDED AFTER THE CONFIRMATION.

	MFG-RegretNet		Uniform split		CSRA, $1/(N-k)$		CSRA, declared†	
α, seed	ϵ	d	ϵ	d	ϵ	d	ϵ	d
0.1, 24	23.90	24.16	14.48	24.50	22.57	26.18	20.98	25.40
0.1, 25	24.06	24.12	23.94	24.10	23.05	23.02	22.33	22.62
0.5, 24	29.16	28.92	26.86	29.23	26.21	28.98	26.91	28.48
0.5, 25	28.87	28.81	29.54	28.88	28.59	27.62	29.14	28.59

(Algorithm 2) for T_{FL}=100 rounds in 18 settings: MNIST with $B \in \{10, 30, 100\}$, CIFAR-10 with $B \in \{30, 100\}$, and both datasets with a purchased privacy of $\bar{\epsilon} \in \{0.5, 0.8\}$ per client and round, each with $\alpha \in \{0.5, 0.1\}$; it uses seeds 231–233, which no development run used, and the official test sets. In the purchased-privacy settings, the budget of every comparator and the purchase T of MFG-RegretNet (Sec. VII-A) are calibrated to a mean release within 0.5% of $\bar{\epsilon}$, so these settings test the selection and the split. Within a seed, all mechanisms share the partition, bids, initial model, and noise streams, and the clients run DP-SGD calibrated to their releases, without a cumulative cap. The decision compares MFG-RegretNet with the best of the ten external IR-feasible mechanisms of each setting by the final accuracy $\mathcal{A}_{\mathrm{final}}$, the mean test accuracy of rounds 96–100: averaged over the settings of a dataset, the margin must be positive in every seed on CIFAR-10 and at least −0.3 points on MNIST.

The decision is met (Table VI): MFG-RegretNet exceeds the best external mechanism by 2.24, 1.35, and 1.51 points on CIFAR-10 in the three seeds (mean 1.70) and by 0.04 points on MNIST (seed margins −0.97 to +0.85). By the three-seed mean, it is ahead in all eight CIFAR-10 settings, by more at matched privacy (2.08 points) than with fixed budgets (1.32), and in five of the ten MNIST settings, within 0.4 points in the others. In every purchased-privacy setting, the mean cumulative ϵ of its clients is lower than under the best external mechanism; with fixed budgets on CIFAR-10, it is higher than under CSRA, the best external mechanism there (5.0–5.8 against 2.9–5.5). Figs. 9 and 10 show the accuracy per round.

A control registered before its runs (Table VII) attributes most of the gain over a uniform split of the same privacy to the budget weights that the allocation implies, so the accuracy results hold for budget-weighted aggregation.

C. Discussion of Results and Limitations

Summary. MFG-RegretNet keeps its incentive properties in measurement and by construction: no offered contract has negative utility, its regret with verified data sizes stays below 10^{-5} in every scenario, and an online attacker or a coalition within its caps gains at most about 1%. It buys a roughly fixed amount of privacy at close to the cap price, so its high welfare is a transfer, and its exact payments make it the slowest mechanism. Its accuracy gains, clearest on CIFAR-10, hold for budget-weighted aggregation: with data-size weights, a uniform split of the same privacy does as well on development data. On MNIST the strongest mechanisms lie within one point of each other. Its data-aligned split serves high-sensitivity clients but excludes clients with a very small data share and does not favor cheap privacy; service or compensation floors can be added. *Heterogeneity.* In the heterogeneous valuation scenarios of Supplementary Table S7, the regret with verified data sizes stays below 10^{-5} and no offered contract has negative utility. Without verification, in the bimodal scenario with κ=3, two of 50,000 clients at B=50 and one at B=100, each with a data share just above τ, could gain up to 0.057 by underreporting their size to leave the selection and collect a larger dividend, and nothing without the dividend; verified

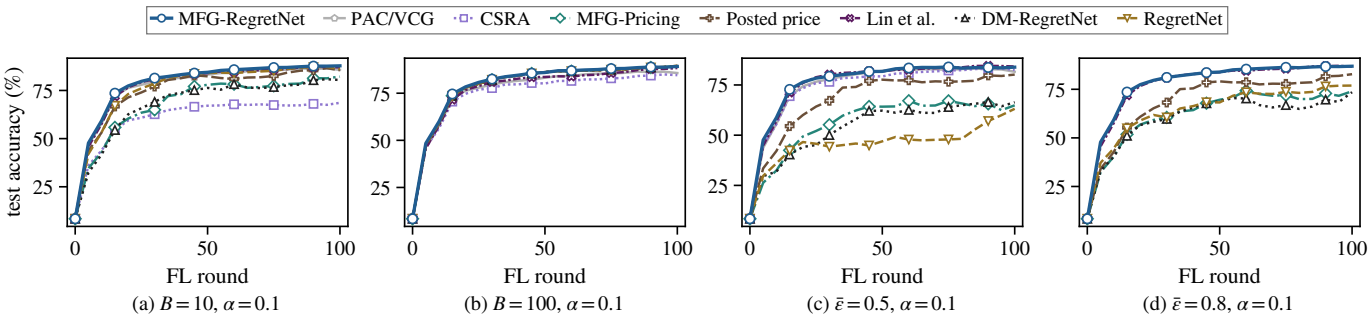


Fig. 9. RQ4: test accuracy per FL round on MNIST in four settings with α=0.1, mean over the three confirmation seeds. For display, the curves take every fifth round and, as the last point, $\mathcal{A}_{\text{final}}$ (the mean of rounds 96–100), smoothed by a centered moving average over $\pm$10 rounds that narrows at the ends; Supplementary Figs. S2 and S3 show every setting, smoothed in the same way. A mechanism that no budget brings to $\bar{\epsilon}$ in every seed has no curve.

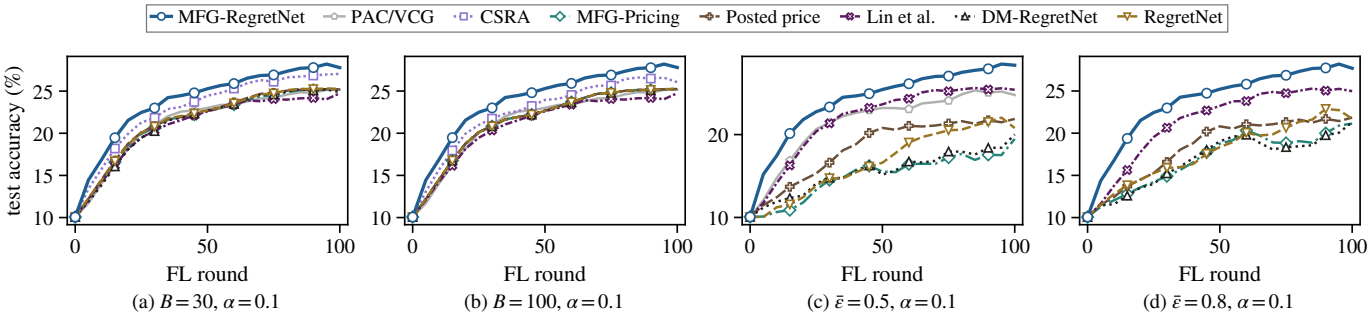


Fig. 10. RQ4: test accuracy per FL round on CIFAR-10, as in Fig. 9. In (d), the best external mechanism, VCG with capped releases, is not drawn.

sizes (Sec. IV-B) exclude this deviation. Three limitations remain. *(L1) Deployment and scale:* all results except the deployment test of Supplementary Sec. S1.J are single-machine simulations with synthetic valuations, and the threshold table and the purchase scale are set for N=10; with many clients of similar size, the selection keeps a single client, so larger populations need thresholds relative to $1/N$ and a purchase scale that grows with N. *(L2) Dynamic privacy:* the accountant tracks cumulative exposure, but each FL round is still an independent auction; a dynamic PAG should price the remaining budget as a mechanism state. *(L3) Trust:* the mechanism assumes a semi-trusted AB, RQ2 measures a semi-honest secure broker, and Proposition 2 covers unilateral deviations; malicious brokers and coalitions beyond the tested modes need audits or coalition-proof extensions.

IX. Conclusion

This paper develops a privacy-trading framework for federated learning incentive design that treats the personalized privacy budget ϵ_i as a *tradable commodity* rather than a fixed system constraint. We formalized the Privacy Auction Game with DSIC, IR, and budget feasibility guarantees for the analytical PAC mechanism (Theorem 1) and a characterization of truthful two-dimensional reporting (Proposition 1), and used a mean-field approximation in which each client responds to the population law; the analytical mechanism converges to its closed-form mean-field limit at rate $\mathcal{O}_P(N^{-1/2})$ (Theorem 4). Under budget-proportional aggregation, the FL error is bounded by a data-alignment term plus a DP noise term (Lemma 1). MFG-RegretNet sets how much privacy to buy from the mean-field statistic of the bids and splits it in proportion to the data of the clients whose data share reaches a public threshold; its monotone networks and Myerson payments make it DSIC, IR, and budget-feasible for every parameter value (Proposition 2), with regret below $\bar{v}\bar{\epsilon}/n$ on every profile (Corollary 2), and its data dividend returns part of the unspent budget to the clients without changing what is bought. In a confirmation fixed before its runs and with budget-weighted aggregation, it exceeds the best external mechanism of each setting on CIFAR-10 in every seed, by 1.70 points on average, and is on par with it on MNIST; a registered control traces its gain over a uniform split of the same privacy to these aggregation weights. Its measured regret stays below 10^{-5}, and because it pays close to the cap price, the clients' surplus comes mainly from the dividend.

References

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.* PMLR, Apr. 2017, pp. 1273–1282.

[2] J. Wu and W. Zhang, "On the security of verifiable and oblivious secure aggregation for privacy-preserving federated learning," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 5, pp. 4324–4326, 2024.

[3] K. Gu, W. Zhao, J. Tan, Z. Yang, X. Li, and W. Jia, "Two-dimensional privacy-preserving federated learning scheme against poisoning attacks," *IEEE Trans. Dependable Secure Comput.*, 2026.

[4] J. Wu, F. Luo, T. Sun, H. Wang, and W. Zhang, "Privacy-preserving federated learning scheme with mitigating model poisoning attacks: Vulnerabilities and countermeasures," *IEEE Trans. Dependable Secure Comput.*, 2025.

[5] H. Yang, M. Ge, D. Xue, K. Xiang, H. Li, and R. Lu, "Gradient leakage attacks in federated learning: Research frontiers, taxonomy and future directions," *IEEE Netw.*, 2023.

[6] R. C. Geyer, T. Klein, and M. Nabi, "Differentially private federated learning: A client level perspective," *arXiv preprint arXiv:1712.07557*, 2017.

[7] Q. Li, Z. Liu, Q. Li, and K. Xu, "martFL: Enabling Utility-Driven Data Marketplace with a Robust and Verifiable Federated Learning Architecture," in *Proc. 2023 ACM SIGSAC Conf. Comput. Commun. Secur.* ACM, Nov. 2023, pp. 1496–1510.

[8] S. Zheng, Y. Cao, M. Yoshikawa, H. Li, and Q. Yan, "Fl-market: Trading private models in federated learning," in *2022 IEEE Int. Conf. Big Data (Big Data)*. IEEE, 2022, pp. 1525–1534.

[9] T. Mai, H. Yao, J. Xu, N. Zhang, Q. Liu, and S. Guo, "Automatic Double-Auction Mechanism for Federated Learning Service Market in Internet of Things," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 5, pp. 3123–3135, Sept. 2022.

[10] X. Lin, J. Wu, J. Li, C. Sang, S. Hu, and M. J. Deen, "Heterogeneous differential-private federated learning: Trading privacy for utility truthfully," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 6, pp. 5113–5129, 2023.

[11] X. Wu, L. Zhang, H. Zhou, J. Tao, Y. Gu, and S. Shen, "Grouping competition-based privacy allocation mechanism for federated learning," *Knowl.-Based Syst.*, vol. 332, p. 114906, 2026.

[12] K. Daskalakis, *The complexity of Nash equilibria*. University of California, Berkeley, 2008.

[13] R. B. Myerson, "Optimal auction design," *Math. Oper. Res.*, vol. 6, no. 1, pp. 58–73, 1981.

[14] A. Ghosh and A. Roth, "Selling privacy at auction," in *Proc. 12th ACM Conf. Electron. Commerce*, 2011, pp. 199–208.

[15] P. Dütting, Z. Feng, H. Narasimhan, D. C. Parkes, and S. S. Ravindranath, "Optimal Auctions through Deep Learning: Advances in Differentiable Economics," *J. ACM*, vol. 71, no. 1, pp. 1–53, Feb. 2024.

[16] L. K. Fleischer and Y.-H. Lyu, "Approximately optimal auctions for selling privacy when costs are correlated with data," in *Proc. 13th ACM Conf. Electron. Commerce*, 2012, pp. 568–585.

[17] Y. Yang, M. Hu, Y. Zhou, X. Liu, and D. Wu, "CSRA: Robust Incentive Mechanism Design for Differentially Private Federated Learning," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 892–906, 2024.

[18] K. Sun, J. Wu, and J. Li, "Reputation-aware incentive mechanism of federated learning: A mean field game approach," in *2024 9th IEEE Int. Conf. Smart Cloud (SmartCloud)*. IEEE, 2024, pp. 48–53.

[19] J. Hu-Bolz, M. Reed, K. Zhang, Z. Liu, and J. Hu, "Federated data acquisition market: Architecture and a mean-field based data pricing strategy," *High-Confidence Comput.*, vol. 5, no. 1, p. 100232, 2025.

[20] S. Jiang, Y. Liu, Z. Wang, Y. Li, H. Chi, and X. Du, "Hybrid makes better: Hybrid differential privacy medical image classification based on federated learning," *IEEE Trans. Dependable Secure Comput.*, 2025.

[21] C. Zhang, J. Weng, J. Weng, Y. Zhong, J.-N. Liu, and C. Deng, "Robust and secure federated learning with verifiable differential privacy," *IEEE Trans. Dependable Secure Comput.*, 2025.

[22] Y. Zhou, X. Liu, Y. Fu, D. Wu, J. H. Wang, and S. Yu, "Optimizing the numbers of queries and replies in convex federated learning with differential privacy," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 6, pp. 4823–4837, 2023.

[23] R. Xu, B. Li, C. Li, J. B. Joshi, S. Ma, and J. Li, "Tapfed: Threshold secure aggregation for privacy-preserving federated learning," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 5, pp. 4309–4323, 2024.

[24] Z. Yi, Y. Jiao, W. Dai, G. Li, H. Wang, and Y. Xu, "A Stackelberg Incentive Mechanism for Wireless Federated Learning With Differential Privacy," *IEEE Wireless Commun. Lett.*, vol. 11, no. 9, 2022.

[25] L. Xie, Z. Su, Y. Wang, N. Chen, Y. Liu, R. Wang, X. Liu, D. Liu, and H. Zhang, "A privacy-preserving incentive scheme for uav-aided federated learning: A contract method with prospect theory," *IEEE Trans. Dependable Secure Comput.*, 2025.

[26] M. B. Singh, H. Singh, and A. Pratap, "Stable Matching based Revenue Maximization for Federated Learning in UAV-Assisted WBANs," *IEEE Trans. Serv. Comput.*, pp. 1–12, 2024.

[27] Y. Zhan, P. Li, Z. Qu, D. Zeng, and S. Guo, "A learning-based incentive mechanism for federated learning," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6360–6368, 2020.

[28] X. Tang, H. Yu, and X. Li, "A reinforcement learning-based bidding strategy for data consumers in auction-based federated learning," *Adv. Neural Inf. Process. Syst.*, vol. 38, pp. 82 666–82 684, 2026.

[29] A. Ghosh, K. Ligett, A. Roth, and G. Schoenebeck, "Buying private data without verification," in *Proc. 15th ACM Conf. Econ. Comput.*, 2014, pp. 931–948.

[30] T. Li, M. Sanjabi, A. Beirami, and V. Smith, "Privacy-aware federated learning with heterogeneous data," in *Int. Conf. Learn. Represent.*, 2022.

[31] M. Noble, A. Bellet, and A. Dieuleveut, "Differentially private federated learning on heterogeneous data," in *Int. Conf. Artif. Intell. Statist.* PMLR, 2022, pp. 10 110–10 145.

[32] R. Xu, Y.-J. A. Zhang, and J. Huang, "JSAM: Privacy straggler-resilient joint client selection and incentive mechanism design in differentially private federated learning," *arXiv preprint arXiv:2602.21844*, 2026.

[33] Z. Jorgensen, T. Yu, and G. Cormode, "Conservative or liberal? personalized differential privacy," in *2015 IEEE 31St Int. Conf. Data Eng.* IEEE, 2015, pp. 1023–1034.

[34] Y. Hu, Y. Wang, J. Lou, W. Liang, R. Wu, W. Wang, X. Li, J. Liu, and Z. Qin, "Privacy risks of federated knowledge graph embedding: New membership inference attacks and personalized differential privacy defense," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 3, pp. 2788–2805, 2024.

[35] C. Dwork, A. Roth, *et al.*, "The algorithmic foundations of differential privacy," *Found. Trends Theor. Comput. Sci.*, vol. 9, no. 3–4, pp. 211–407, 2014.

[36] I. Mironov, "Rényi differential privacy," in *Proc. IEEE 30th Comput. Secur. Found. Symp. (CSF)*, 2017, pp. 263–275.

[37] M. Zhang, F. Beltran, and J. Liu, "Selling Data at an Auction under Privacy Constraints," in *Proc. 36th Conf. Uncertainty Artif. Intell. (UAI)*. PMLR, Aug. 2020, pp. 669–678.

[38] A. Acquisti, L. K. John, and G. Loewenstein, "What is privacy worth?" *J. Legal Stud.*, vol. 42, no. 2, pp. 249–274, 2013.

[39] W. Cheng, Y. Zou, J. Xu, and W. Liu, "Dynamic games for social model training service market via federated learning approach," *IEEE Trans. Comput. Social Syst.*, vol. 9, no. 1, pp. 64–75, 2021.

[40] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-iid data," *arXiv preprint arXiv:1806.00582*, 2018.

[41] J.-M. Lasry and P.-L. Lions, "Mean field games," *Jpn. J. Math.*, vol. 2, no. 1, pp. 229–260, 2007.

[42] M. Huang, R. P. Malhamé, and P. E. Caines, "Large population stochastic dynamic games," *Commun. Inf. Syst.*, vol. 6, no. 3, pp. 221–252, 2006.

[43] N. Fournier and A. Guillin, "On the rate of convergence in wasserstein distance of the empirical measure," *Probab. Theory Relat. Fields*, vol. 162, no. 3, pp. 707–738, 2015.

[44] M. Ajtai, J. Komlós, and G. Tusnády, "On optimal matchings," *Combinatorica*, vol. 4, no. 4, pp. 259–264, 1984.

[45] P. Massart, "The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality," *Ann. Probab.*, vol. 18, no. 3, pp. 1269–1283, 1990.

[46] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[47] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009, technical Report.

[48] W. Vickrey, "Counterspeculation, auctions, and competitive sealed bids," *J. Finance*, vol. 16, no. 1, pp. 8–37, 1961.

[49] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Theory Cryptogr. Conf.* Springer, 2016, pp. 635–658.

[50] P. Kairouz, S. Oh, and P. Viswanath, "The composition theorem for differential privacy," in *Int. Conf. Mach. Learn.* PMLR, 2015, pp. 1376–1385.

[51] B. Knott, S. Venkataraman, A. Hannun, S. Sengupta, M. Ibrahim, and L. van der Maaten, "CrypTen: Secure multi-party computation meets machine learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021.

[52] P. Yuhala, M. Paper, T. Zerbib, P. Felber, V. Schiavoni, and A. Tchana, "SGX switchless calls made configless," in *Proc. 53rd Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, 2023.